Copyrighted Materials
Copyright © 2002 Oxford University Press Retrieved from www.knovel.com

4

Vibrations Involving More Than One Degree of Freedom

4.1 INTRODUCTION

Although some systems can be analyzed from a single-degree-of-freedom (SDOF) standpoint, there are a great many situations for which this viewpoint is simply insufficient. In one example, to be studied in some detail in this chapter, a designer purposely puts an additional degree of freedom in an existing system to modify the original system's vibrational response. The devices that do this are known as vibration absorbers. Automobile designers, when trying to obtain good handling characteristics for their cars, routinely have to include more than just one degree of freedom to capture the various motions the car exhibits, such as yaw, heave, pitch, and roll. Even the simplest model of a steam turbine should include at least two degrees of freedom, while reasonable models for large space structures routinely involve hundreds.

The fundamental difference between a multi-degree-of-freedom (MDOF) system and one with a single degree of freedom is that in the first case we've got additional masses to worry about. The important consequence of this fact is that a sinusoidal response of the MDOF system will automatically imply a relationship between the motions of the various masses in the structure. For instance, the motion of one mass in a 2 DOF system might be described by a cosine and the motion of the other

by a cosine having twice the amplitude of the first. Thus, to describe the motion, we should concern ourselves with the *relative* motion between the two masses, since the fundamental characteristic (cosine oscillations) is the same for both masses. In the case just mentioned, this would mean noting that one mass moves twice as much as the other. This idea of relative motion is a crucial one, and since it underlies most of the remaining chapters, we'll study it in some detail.

4.2 FREE RESPONSE—UNDAMPED SYSTEM

We'll start by analyzing the system shown in Figure 4.1. External forcings and damping have been excluded for clarity and will be examined later in the chapter.

Since 2 DOF problems are easily visualized, can be analytically solved, and can be experimentally demonstrated by using a simple spring with some masses, we'll focus on them when introducing and discussing multi-degree-of-freedom characteristics. The findings, however, will be equally applicable to problems with any desired number of degrees of freedom.

As we can see from the free body diagram of Figure 4.1*b*, the equations of motion for this system can be written as

$$m_1\ddot{x}_1 + (k_1 + k_2)x_1 - k_2x_2 = 0 \tag{4.2.1}$$

and

$$m_2\ddot{x}_2 + (k_2 + k_3)x_2 - k_2x_1 = 0 \tag{4.2.2}$$

At this point we'll make the most important transition of the book. Up until now, everything we've been dealing with has involved scalar quantities. But we now have the opportunity to move beyond this stage and into the vector/matrix world, because the foregoing equations can just as easily be

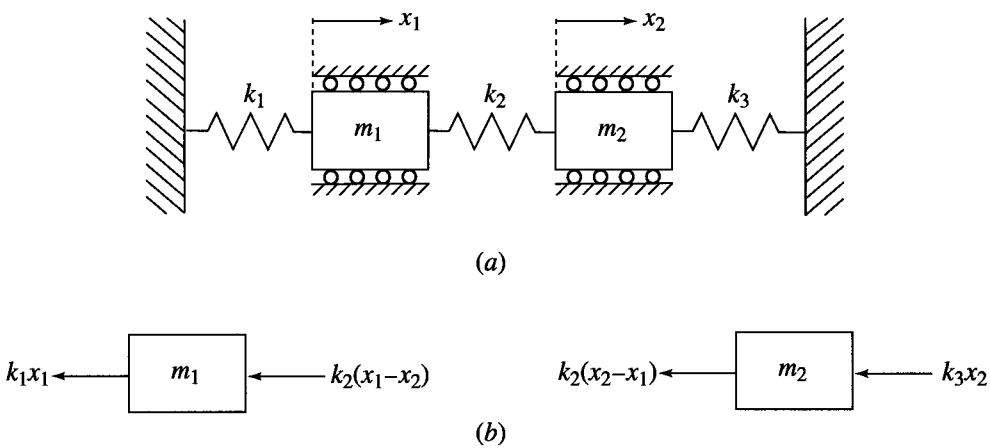


Figure 4.1 (*a*) 2-DOF system and (*b*) free body diagram

reexpressed as

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4.2.3}$$

If we define

$$[M] \equiv \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}, \ [K] \equiv \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix}, \ X \equiv \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}, \text{ and } \ O \equiv \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4.2.4}$$

then we can express (4.2.3) as

$$[M]\ddot{X} + [K]X = O \tag{4.2.5}$$

Thus we have a form that is, on the surface, identical to that of our SDOF oscillator. The difference now is that our "masses" and "springs" are matrices and our dependent variable is a vector, not a scalar.

The definitions just given follow the convention that will be used throughout the rest of the book. Lowercase letters will be used to represent scalar quantities. Thus, when you see something like b, you'll know it is a scalar. Vectors are denoted by an uppercase letter, such as X or Y. Finally, matrices are indicated by an uppercase letter surrounded by square brackets, such as $[A]$. When referring to the entries in a vector or matrix, the letter of the vector/matrix in lowercase form will be used, along with an added subscript. Thus $a_{1,2}$ is the entry in the first row and second columns of $[A]$. As a further example, the first row, first column entry in $[K]$ (defined in 4.2.4) is $k_{1,1}$ and

$$k_{1,1} = k_1 + k_2 \tag{4.2.6}$$

Even though the actual entries are denoted by k, you can tell the difference between them and the symbol indicating a particular matrix entry because the matrix entry will show the row and column number, by means of two subscripts, separated by a comma. The only exceptions to this convention are certain parameters that are traditionally written in uppercase letters (such as Young's modulus, E). It will be clear from the context that these symbols represent parameters, not vectors.

Now that we've entered the matrix/vector world, we're not going to leave it. For the rest of the book, we'll use matrix equations in one form or another. More specifically, we'll use (4.2.5) quite a bit, and thus all the analyses and conclusions we draw in this chapter will be very applicable to the problems we'll encounter in the future.

Before analyzing (4.2.5) any further, you should see what you can recall from your class in linear algebra. Almost certainly, part of what you learned included an analysis of the matrix equation

$$[A(\lambda)]X = O \tag{4.2.7}$$

It's also possible that you dealt with a different form of this equation, namely

$$[A]X = \lambda X \tag{4.2.8}$$

In either case, you'll recall that these were called *eigenvalue problems* and that the solution involved finding both the eigenvalues and eigenvectors for the system. What's important is that (4.2.5) will lead to just such an eigenvalue problem.

To see this, let's presume a solution to (4.2.5) in the form

$$X(t) = \bar{X}e^{i\omega t} = \begin{Bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{Bmatrix} e^{i\omega t} \tag{4.2.9}$$

We know that this is an appropriate form because, as we've stated before, we should always expect linear, constant coefficient, ordinary differential equations to support exponential solutions. We've got three unknowns to deal with, $\bar{x}_1$, $\bar{x}_2$, and ω. We chose $e^{i\omega t}$ instead of $e^{\lambda t}$ because of our earlier observation that a spring-mass system supports sinusoidal responses rather than real exponential solutions. However, you should realize that we could use $e^{\lambda t}$ just as easily; it won't affect the final results. You'll notice that we've got three unknowns and only two equations. Thus we know that we won't get unique numerical solutions for $\bar{x}_1$, $\bar{x}_2$, and ω. This is fine, as we'll see shortly.

To solve (4.2.5), we'll use our assumed solution, (4.2.9), differentiate it twice (for the $\ddot{X}$ term), substitute it in, and find (factoring out the $e^{i\omega t}$ term)

$$-\omega^2[M]\bar{X} + [K]\bar{X} = O \tag{4.2.10}$$

We can now factor out the $\bar{X}$ term to obtain

$$\left[[K] - \omega^2[M]\right]\bar{X} = O \tag{4.2.11}$$

This looks exactly like (4.2.7), (in which $[A]$ is equal to $([K] - \omega^2[M])$. Thus we have an eigenvalue problem. Although $\bar{X} = O$ solves this equation, we recognize this as the trivial solution—trivial in that $\bar{X} = O$ implies no motion at all. Consequently we can ignore this solution as being of no interest. We're left with asking when an equation of the form

$$[A]\bar{X} = O \tag{4.2.12}$$

has a solution for finite $\bar{X}$. The answer is, when $[A]$ is not invertible. And why is this? Well, if $[A]$ was invertible, then we could compute its inverse and premultiply (4.2.12) by $[A]^{-1}$ to obtain

$$[A]^{-1}[A]\bar{X} = [A]^{-1}O \tag{4.2.13}$$

or

$$\bar{X} = O \tag{4.2.14}$$

Thus, if $[A]$ was invertible, we'd be left with the trivial solution. Therefore, if we want a more interesting solution, $[A]$ must not be invertible. And for a matrix to not be invertible, its determinant must be equal to zero. This means that our requirement for a solution is given by

$$\det\left([K] - \omega^2[M]\right) = 0 \tag{4.2.15}$$

Thus

$$\det\left(\begin{bmatrix} k_1 + k_2 - \omega^2 m_1 & -k_2 \\ -k_2 & k_2 + k_3 - \omega^2 m_2 \end{bmatrix}\right) = 0 \tag{4.2.16}$$

If we evaluate this determinant, we'll obtain:

$$(k_1 + k_2 - \omega^2 m_1)(k_2 + k_3 - \omega^2 m_2) - k_2^2 = 0 \tag{4.2.17}$$

Expanding this out and grouping the powers of ω leads to

$$m_1 m_2 \omega^4 - \omega^2((k_2 + k_3)m_1 + (k_1 + k_2)m_2) + k_1 k_2 + k_1 k_3 + k_2 k_3 = 0 \tag{4.2.18}$$

To provide further insight, let's assign specific values to the springs and masses. We'll let m_1 and m_2 each have a mass of 1 kg and let the k_i all be equal to 4 N/m. Under these conditions, (4.2.18) becomes

$$\omega^4 - 16\omega^2 + 48 = 0 \tag{4.2.19}$$

This polynomial is called the *characteristic polynomial*, and, when set equal to zero, gives us the *characteristic equation* for the system. Although we'll refer to the roots of this polynomial as our natural frequencies (or damped frequencies of oscillation for the case with damping), we'd call them the "poles" of the system if we were looking at it from a controls perspective. So for those of you who have studied (or plan to study) control theory, remember that the poles of a system and an undamped system's natural frequencies are the same thing.

Recognizing that this equation isn't really a full fledged, fourth-order equation but is rather quadratic in ω^2, we can solve for ω^2 straightforwardly:

$$\omega_{1,2}^2 = \frac{1}{2}(16 \pm \sqrt{16^2 - (4)(48)}) \tag{4.2.20}$$

or

$$\omega_1^2 = 4 \text{ (rad/s)}^2 \tag{4.2.21}$$

and

$$\omega_2^2 = 12 \text{ (rad/s)}^2 \tag{4.2.22}$$

Labeling the natural frequencies in ascending order ($\omega_1 \leq \omega_2 \leq \omega_3 \cdots$) is the typical convention and the one we'll follow. Just as in the SDOF case, we don't need to worry overly much about obtaining four values for ω when we take the square roots of ω_1^2 and ω_2^2, since these will combine to give us sine and cosine solutions just as they did before. These two values for ω^2 are eigenvalues for our problem, and their square roots are equal to the two possible frequencies of oscillation for the system. Note that for the single-degree-of-freedom case the only possible free vibration frequency was ω_n. Now we have two frequencies.

Once we've found the two solutions for ω^2, we're left with determining $\bar{x}_1$ and $\bar{x}_2$. To go further with our solution, we'll revisit (4.2.11). With ω^2 now known, we simply need to specify which

of the two solutions we want to use. To start, we'll look at the lower frequency solution, that is, $\omega_1^2 = 4\ (\text{rad/s})^2$. Substituting this value for ω^2 and inserting the appropriate values for m_i and k_i gives us

$$-4\left[\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix}\right]\begin{Bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4.2.23}$$

or more simply

$$\begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix}\begin{Bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4.2.24}$$

Both equations in (4.2.24) indicate the same thing, namely, that $\bar{x}_1$ must be equal to $\bar{x}_2$. Getting the same equation makes sense: it implies that the two rows of the matrix are not independent, something that has to be true if the matrix is to be noninvertible.

We're now in a position to write down one of the solutions to our problem. Based on the results so far, we know that one possible solution is

$$X(t) = a_1 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} e^{2it} + a_2 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} e^{-2it} \tag{4.2.25}$$

where a_1 and a_2 are as yet undetermined constants we could solve for if we were given specific initial conditions.

A more accessible (and equivalent) form is to use a trigonometric representation instead of the exponential form shown in (4.2.25):

$$X(t) = b_1 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \cos(2t - \phi_1) \tag{4.2.26}$$

Since we've included a phase shift, this representation can include pure sines and cosines as well as a mix of the two, as discussed earlier.

Equation (4.2.26) represents one of the fundamental responses of the system. Physically, both masses are locked together in their motions: when one moves to the right, so does the other, and to the same degree. You'll also note that the frequency of this oscillation, 2 rad/s, is the same as the natural frequency we'd find from analyzing one of the springs attached to one of the masses. The reason for this is not too difficult to see. Imagine that the middle spring has been removed and you move both masses the same amount to the left and then release them. Since they're now simply two independent, single-degree-of-freedom systems, they'll oscillate at their natural frequencies. And since the masses and springs are identical, the natural frequency of oscillation will be the same for both. Thus the masses will move in synchronism with each other and the distance between them will stay fixed. Therefore it doesn't matter whether there is a spring between them or not; since the relative distance between them stays constant, the middle spring isn't stretched or compressed anyway. This is a consequence of both the left and

right spring-mass systems having the same natural frequencies. Whenever you have this situation, the first natural frequency of the 2 DOF system will be the same as the individual SDOF systems.

We call the vector $(1 \;\; 1)^T$ shown in (4.2.25) and (4.2.26) an *eigenvector* of our system. You'll also see it called modal vector, eigenmode, natural mode, and sometimes simply mode. It indicates the relative displacements (and later on will include velocities) of the different masses in the system. Since the system has no damping, the oscillations go on forever and, if the system is initially displaced into the first eigenmode, it remains there. Note that the amplitude of the eigenvector is arbitrary, just as the amplitude was arbitrary in the SDOF case. So the really important thing about this eigenmode is that both entries are the same.

We can now solve for the second mode of the system exactly as we solved for the first. In this case ω^2 is 12 rad/s^2 and our equation governing the second mode is given by

$$\begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix} \begin{Bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4.2.27}$$

and our solution is

$$X(t) = a_3 \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} e^{2\sqrt{3}it} + a_4 \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} e^{-2\sqrt{3}it} \tag{4.2.28}$$

or, more familiarly

$$X(t) = b_2 \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \cos(2\sqrt{3}t - \phi_2). \tag{4.2.29}$$

Using X_i to identify our eigenvectors

$$X_1 \equiv \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}, \quad X_2 \equiv \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \tag{4.2.30}$$

we can write our total solution for this unforced, 2 DOF oscillator as

$$X(t) = b_1 X_1 \cos(\omega_1 t - \phi_1) + b_2 X_2 \cos(\omega_2 t - \phi_2) \tag{4.2.31}$$

where ω_1 is the first natural frequency of the system (and is equal to 2 rad/s) and ω_2 is the second natural frequency (equaling $2\sqrt{3}$ rad/s).

As a concrete example of how we would match specific initial conditions, consider the response of the system to the initial conditions

$$x_1(0) = 1 \text{ mm}, \;\; \dot{x}_1(0) = 0, \;\; x_2(0) = 0, \;\; \dot{x}_2(0) = 0 \tag{4.2.32}$$

Note that this initial deflection does not match either the first or the second eigenvector's shape. Thus we would predict the involvement of both eigenvectors in the subsequent motion. And we'd be right,

because the solution for this case is

$$\phi_1 = 0, \quad \phi_2 = 0, \quad b_1 = .5 \text{ mm}, \quad b_2 = .5 \text{ mm} \tag{4.2.33}$$

The total response can be written as

$$X(t) = X_1 \cos(2t) + X_2 \cos(2\sqrt{3}t) \tag{4.2.34}$$

It's important to realize that even if the initial deflections of the two masses match only one particular eigenvector, both eigenvectors might be involved in the response because of the initial velocities. For instance, the initial conditions

$$x_1(0) = 1, \; x_2(0) = 1, \; \dot{x}_1(0) = 2\sqrt{3}, \; \dot{x}_2(0) = \sqrt{3} \tag{4.2.35}$$

lead to the solution

$$X(t) = X_1 \cos(2t) + X_2 \cos(2\sqrt{3}t - \tfrac{\pi}{2}) \tag{4.2.36}$$

even though no trace of the second eigenvector showed up in the initial displacements. Thus both velocities and displacements must correspond to those supported by a particular eigenvector for that to be the sole contributor to the response.

Although we can solve 2×2 problems by hand, it's more convenient to let the computer do the work, especially as we move into more complicated problems. MATLAB is a very nice piece of software that allows problems such as ours to be solved quickly and easily. There are other software packages out there as well but MATLAB is very widely used across the country and I'll be using it in this book.

MATLAB has two basic routines that handle eigenvalue problems, both called "eig." If you type

```
eig([A])  (cr)
```

where (cr) indicates a carriage return, MATLAB will determine the solutions to the problem:

$$\dot{X} = [A]X \tag{4.2.37}$$

If you type

```
eig([A], [B])  (cr)
```

then MATLAB will solve

$$[B]\dot{X} = [A]X \tag{4.2.38}$$

Note that within MATLAB you can call your matrix m or M or $[M]$; it won't matter. Since typing M is quicker than typing $[M]$, it'll be easier to forget the brackets when in MATLAB. However, to remain consistent, when referring to a matrix in the discussion we'll use the bracket notation ($[M]$) but from now on, when giving directions for MATLAB use we'll use the simpler M form.

In MATLAB, you can solve for the eigenvectors and eigenvalues of a spring-mass system by typing

$$[\mathrm{v}, \mathrm{d}] = \mathtt{eig}(\mathrm{K}, \mathrm{M})$$

where K is the system's spring matrix and M is the mass matrix. MATLAB doesn't care (within limits) how big your system is. If M and K are $n \times n$ matrices, then v will be a matrix whose columns will be the system's eigenvectors, and d will be a diagonal matrix having the ω_i^2 values on the diagonal. Since these matrices correspond to each other, the second column of (v), for example, will be the eigenvector associated with the d(2, 2) entry. Note that MATLAB will not order the natural frequency entries in any particular fashion; i.e., they won't appear from smallest to largest.

Example 4.1

Now that you've seen what to expect in 2 DOF systems, you can go ahead and use your spring to run your own 2 DOF experiment. First, you'll need to cut the spring into two pieces. Next, you'll need two masses. The more symmetrics they are, the better, since you'll want the motions to stay purely translational; rocking due to imbalances will just make the experiments harder.

You'll be doing two things in this experiment—experimentally verifying the existence of two eigenvectors (modes) and analytically predicting their natural frequencies from a couple of SDOF tests. Even more amazingly, we won't even need to know the actual mass or spring constants. The only additional item you'll need is a watch with a stopwatch function.

To start, attach one of the masses to one spring and the other to the other spring. To minimize confusion, we'll call the mass and spring of the first set m_1 and k_1, and the second set will be denoted by m_2 and k_2. Next, hold the free end of the spring against a fixed surface (like a door frame), pull down on the mass, and let go. The mass will then bob up and down. If you start your stopwatch at a high point of the motion and time the motion for five or ten cycles, you can calculate a pretty accurate estimate of the oscillation period by dividing the total time by the number of cycles that occurred. If you call this period t_1, then the $\frac{k_1}{m_1}$ is equal to $\left(\frac{2\pi}{t_1}\right)^2$. You should then repeat this procedure with m_2 and k_2, leading to an estimate of $\frac{k_2}{m_2}$. Finally, hook m_1 and k_2 together and determine $\frac{k_2}{m_1}$. What you'll now do is attach m_1 to the free end of k_2, as shown in Figure 4.2. If you hold the free end of k_1, you'll have a 2 DOF system (for vertical motions).

Getting this system to exhibit its first and second eigenmodes isn't difficult. Simply start moving your hand up and down as you did in the SDOF experiments and try to get a resonant response. If you start at a low frequency and slowly increase it, you'll reach a frequency for which very small motions of your hand produce large, in-phase motions of the masses (they both move in the same direction at all times). If you then move your hand faster, you'll reach a point at which the masses move in opposition to each other (one moves up while the other moves down). At this point your hand is oscillating at the system's second natural frequency. Because the damping is so low, the individual mode shapes should be quite apparent. Once the oscillations are established, you can hold your hand against a rigid support and watch the modal vibrations. Figure 4.3 shows what you should be seeing.

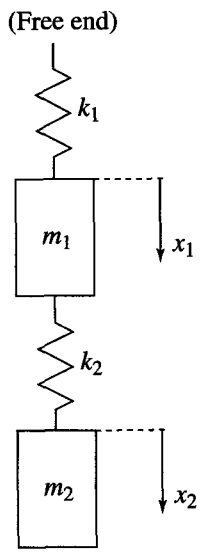


Figure 4.2 Experimental 2 DOF system

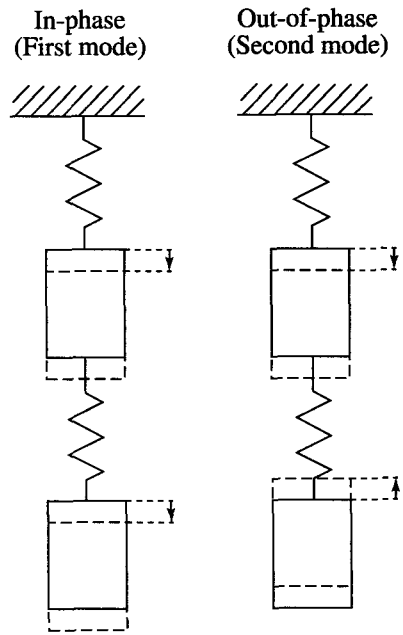


Figure 4.3 Modal behavior of a physical 2 DOF system

All right, now that we've actually seen the eigenmodes in action, it's time to analytically predict what their frequencies should be (and verify the predictions, of course). The equations of motion for our system are given by

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Now how can we progress any further without knowing the particular values of the masses and stiffnesses? By dividing through by the masses! If we divide the first equation by m_1 and the second by m_2, we'll have terms like $\frac{k_1}{m_1}$ and $\frac{k_2}{m_2}$. These are simply the squares of the natural frequencies associated with the simpler single-DOF problems we've already analyzed.

Going ahead and dividing by the masses gives us

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} \frac{k_1}{m_1} + \frac{k_2}{m_1} & -\frac{k_2}{m_1} \\ -\frac{k_2}{m_2} & \frac{k_2}{m_2} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4.2.39}$$

At this point, we have to part company, since I've got no idea what natural frequencies you have. But just for the record, I'll go through what I get when I run this test. For the masses, I used heavy doughnut-shaped pieces of steel (which came out of the weights of a grandfather clock). The periods of oscillation for the two SDOF oscillators were

$$t_1 = .864 \text{ second}$$

and

$$t_2 = .861 \text{ second}$$

which correspond to frequencies of

$$\omega_1 = 7.27 \text{ rad/s}$$

and

$$\omega_2 = 7.30 \text{ rad/s}$$

A ten-cycle average was used when obtaining these values.

If these results are substituted into (4.2.39), the equations become

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 106.2 & -53.3 \\ -53.3 & 53.3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

The characteristic equation is therefore

$$\omega^4 - 159.5\omega^2 + 2820 = 0$$

which has solutions

$$\omega_1 = 4.50 \text{ rad/s}$$

and

$$\omega_2 = 11.80 \text{ rad/s}$$

By exciting the experimental system at resonance, immediately placing the free end of the spring against the top of a door frame, timing 10 oscillations, and then averaging, the following time periods were obtained:

$$t_1 = 1.40 \text{ seconds}$$

and

$$t_2 = .538 \text{ second}$$

These correspond to experimentally derived natural frequencies of

$$\omega_{1_{\text{exp}}} = 4.49 \text{ rad/s}$$

and

$$\omega_{2_{\text{exp}}} = 11.68 \text{ rad/s}$$

Comparing the two sets of results, we see that the predicted first natural frequency was within .2% of the experimental value while the second was off by only 1%.

These excellent results weren't a fluke; hopefully you obtained equally accurate answers during your own test. What this experiment shows is that the modeling we've been talking about is really pretty useful and reflects reality quite well. Of course, you'll rarely be dealing with masses attached to springs in your professional life. But the basic conclusion still holds. The techniques we've learned (and the ones still to come) can do an amazing job of predicting real-world behavior as long as the assumptions that underlie them aren't violated.

Example 4.2

Problem Determine the natural frequencies and eigenvectors for the system shown in Figure 4.1*a*: $m_1 = 2$ kg, $m_2 = 4$ kg, $k_1 = 40$ N/m, $k_2 = 100$ N/m, and $k_3 = 200$ N/m.

Solution The equations of motion for the system are found from performing a force balance:

$$m_1\ddot{x}_1 = -k_1x_1 + k_2(x_2 - x_1)$$
$$m_2\ddot{x}_2 = -k_2(x_2 - x_1) - k_3x_2$$

which can be rewritten as

$$m_1\ddot{x}_1 + (k_1 + k_2)x_1 - k_2x_2 = 0$$
$$m_2\ddot{x}_2 + (k_2 + k_3)x_2 - k_2x_1 = 0$$

Putting this into matrix form and using the given parameter values yields

$$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 140 & -100 \\ -100 & 300 \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

This time I'll use MATLAB. Typing [v,d]=`eig`(K,M) where

$$M = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \text{ and } K = \begin{bmatrix} 140 & -100 \\ -100 & 300 \end{bmatrix}$$

yields

$$\omega_1^2 = 37.056 \;\Rightarrow\; \omega_1 = 6.087 \text{ rad/s and } X_1 = \begin{Bmatrix} .8350 \\ .5502 \end{Bmatrix}$$

$$\omega_2^2 = 107.9 \;\Rightarrow\; \omega_1 = 10.39 \text{ rad/s and } X_1 = \begin{Bmatrix} -.7966 \\ .6045 \end{Bmatrix}$$

Now that we've investigated the eigenvectors a bit, which are a free response characteristic, we'll move on to consider some forced responses and point out how the natural frequencies and eigenvectors affect the solutions.

4.3 FORCED RESPONSE

If we include the applied forces shown in Figure 4.4, our equations of motion become

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix} \tag{4.3.1}$$

For simplicity, assume that the forcing is in the form

$$f_1(t) = \bar{f}_1 \cos(\omega t)$$

$$f_2(t) = \bar{f}_2 \cos(\omega t) \tag{4.3.2}$$

that is, the magnitudes of the two forces can vary but the frequency is the same for each. As in the SDOF problem, when we have a sinusoidal forcing, we should expect a sinusoidal response. And since the problem has no damping, the output should be in phase or 180 degrees out of phase with the input. Of course, we could use complex exponentials to represent the input and output (and we'll do so for the damped case). But for now we'll stick with an explicit cosine forcing, since the math is simpler without any damping and the cosine solution has a more straightforward physical interpretation.

If we let the response be

$$\begin{aligned} x_1(t) &= \bar{x}_1 \cos(\omega t) \\ x_2(t) &= \bar{x}_2 \cos(\omega t) \end{aligned} \tag{4.3.3}$$

and substitute this (along with the forcing) into (4.3.1) then we'll obtain

$$-\omega^2 \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix} \begin{Bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{Bmatrix} = \begin{Bmatrix} \bar{f}_1 \\ \bar{f}_2 \end{Bmatrix} \tag{4.3.4}$$

or

$$\left[[K] - \omega^2 [M]\right] \bar{X} = \bar{F} \tag{4.3.5}$$

where $F \equiv \{\bar{f}_1 \;\; \bar{f}_2\}^T$

Unlike (4.2.11), we now have a nonhomogeneous problem; i.e., we're not dealing with $[A]\bar{X} = O$ but with $[A]\bar{X} = \bar{F}$. And unlike the homogeneous case, $[A]$ must be invertible for a solution to exist. If $[A]^{-1}$ exists, then we'll premultiply by it to find $\bar{X} = [A]^{-1}\bar{F}$. For this 2 DOF example, we can work out the solution explicitly, but for more complex problems we'll need to rely on numerical solutions.

You'll recall from your linear algebra course or from Appendix D that the inverse of a 2×2 matrix is given by

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \frac{1}{(ad - bc)} \tag{4.3.6}$$

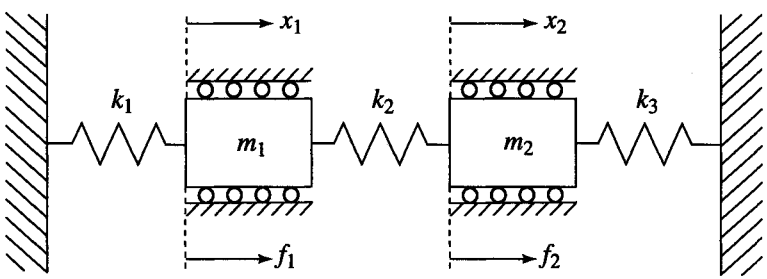


Figure 4.4 Forced 2 DOF spring-mass system

Thus our solution $\bar{X}$ is given by

$$\bar{X} = \frac{1}{\Delta(\omega)} \begin{bmatrix} k_2 + k_3 - m_2\omega^2 & k_2 \\ k_2 & k_1 + k_2 - m_1\omega^2 \end{bmatrix} \begin{Bmatrix} \bar{f}_1 \\ \bar{f}_2 \end{Bmatrix} \tag{4.3.7}$$

where $\Delta(\omega) \equiv (k_1 + k_2 - m_1\omega^2)(k_2 + k_3 - m_2\omega^2) - k_2^2$. The individual components of $\bar{X}$ are given by

$$\bar{x}_1 = \frac{(k_2 + k_3 - m_2\omega^2)\bar{f}_1 + k_2\bar{f}_2}{\Delta(\omega)} \tag{4.3.8}$$

and

$$\bar{x}_2 = \frac{k_2\bar{f}_1 + (k_1 + k_2 - m_1\omega^2)\bar{f}_2}{\Delta(\omega)} \tag{4.3.9}$$

Notice that the denominator for both terms, $\Delta(\omega)$, is the same, and if it is set to zero we have exactly the same equation that we used to solve for the system natural frequencies (4.2.17). Is this a bizarre coincidence? Of course not! This will invariably be the case when we find the forced response of a linear system such as this one. The denominator is always the characteristic polynomial for our system. If you'll recall what was so special about this polynomial, namely, that its roots were the natural frequencies of the system, you'll realize that they play an important role for the forced problem also. Since the characteristic polynomial is equal to zero at the system's natural frequencies, (4.3.8) and (4.3.9) tell us that the denominators go to zero at these frequencies. Thus, if the numerator isn't equal to zero (and it usually isn't), the response will go to infinity. What does this mean? It means that we've now got two resonance conditions, one for each natural frequency. For the SDOF system we had only one resonant frequency. Now, not surprisingly, we've got two. Figure 4.5 shows the response for $\bar{x}_1$ and $\bar{x}_2$ for a representative case (in case you're really interested, the parameters $m_1 = 1$, $m_2 = 1$, $k_1 = 1$, $k_2 = 2$, $k_3 = 2$, $\bar{f}_1 = 2$, and $\bar{f}_2 = -1$ were used). As you can see, the responses for both $\bar{x}_1$ and $\bar{x}_2$ are finite at zero frequency (as in the SDOF problem), go to infinity at the resonant frequencies of the system, and die down to zero as the forcing frequency becomes very large (also like the SDOF problem). Figure 4.5 also shows that the phase response is either 0 or π radians (i.e., a positive or negative response, respectively). Just as for the SDOF problem, the phase shifts π radians after each resonant frequency is passed.

The situation is much the same for higher order systems. For instance, Figure 4.6 shows the response of the first mass in the four-mass system shown in Figure 4.7. With four masses, we now have four natural frequencies, ω_1–ω_4. You can see from Figure 4.6 that the response goes to infinity at each of these frequencies and also that the phase shifts by π radians as we pass these frequencies (i.e., the response switches sign).

Calculating $\bar{x}_1$ and $\bar{x}_2$ with MATLAB is pretty straightforward. If you represented the quantities of interest as $K \equiv [K]$, $M \equiv [M]$, $om \equiv \omega$ and $F \equiv \bar{F}$ then you'd find the desired solution vector

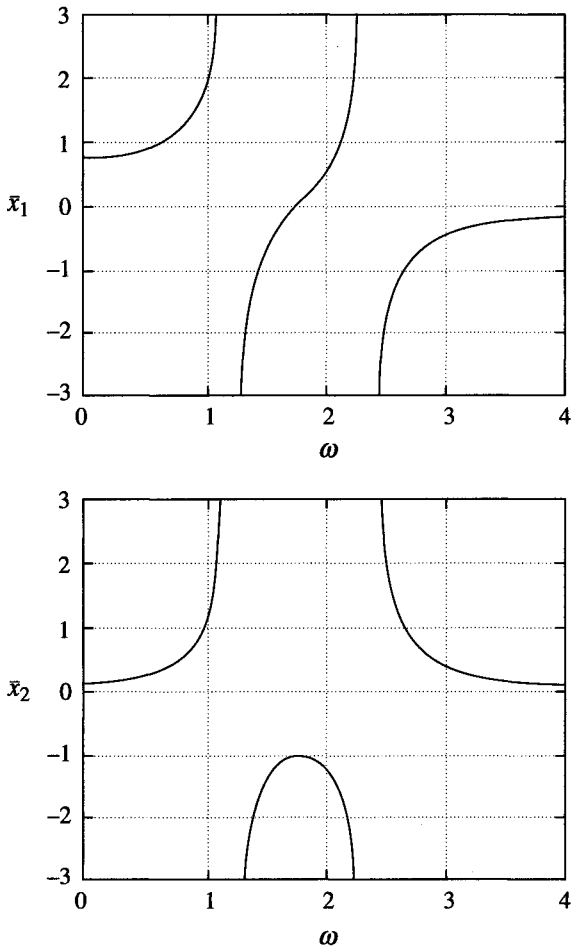


Figure 4.5 $\bar{x}_1$ and $\bar{x}_2$ versus ω

from entering

$$\mathrm{X} = \mathtt{inv}(\mathrm{K} - \mathrm{om} \wedge 2 * \mathrm{M}) * \mathrm{F}$$

Example 4.3

Problem To see how we might use the forced response of an MDOF system, consider Figure 4.8. We can think of this system as approximating an autonomous vehicle, to be used in unmanned explorations of Mars. The two springs represent the front and rear suspensions, and the bar represents the body. We'll assume that the disturbance represents the effects of an unbalanced force in the engine. Our design goal will be to ensure that the vibrations are minimized at the location of the main instrumentation. As a simplification, we'll assume that the inertia of the beam doesn't alter as the instrumentation is moved. Although we'll work out the governing equations for arbitrary values of the various parameters, for our numerical results we'll use $\rho = 50$ kg, $l = 1$ m, $\bar{I} = 16.\bar{6}$ kg·m^2

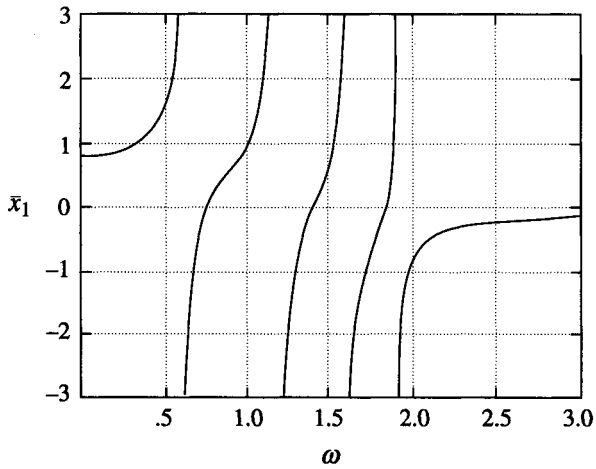


Figure 4.6 Response of the first mass of a four-mass system

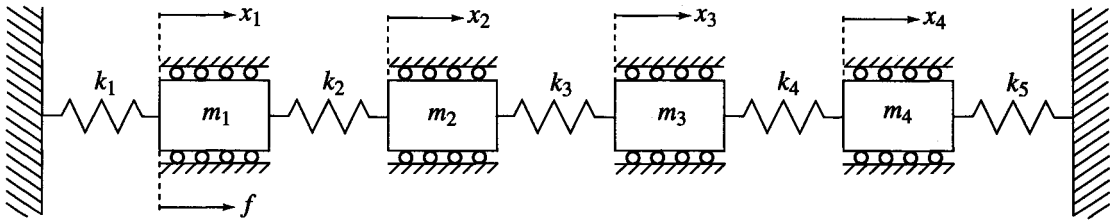


Figure 4.7 Forced, 4 DOF spring-mass system

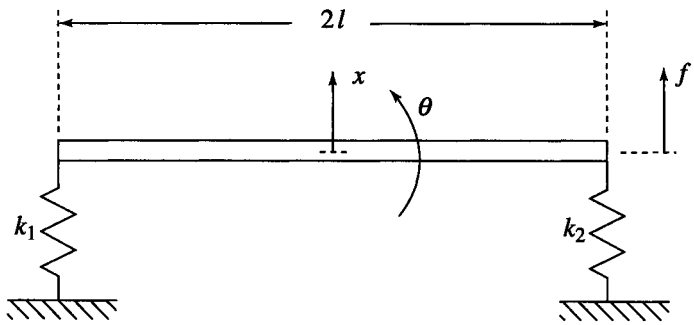


Figure 4.8 2 DOF model of an autonomous vehicle

(mass moment of inertia about the mass center), $k_1 = 30,000$ N/m, $k_2 = 20,000$ N/m, and $f = 100\cos(\omega t)$ N.

Solution To solve this problem, we first need to obtain the system's equations of motion. A convenient set of coordinates is the vertical translation x of the vehicle's center of mass and the rotation θ about that center. Summing moments and forces about the center of mass will

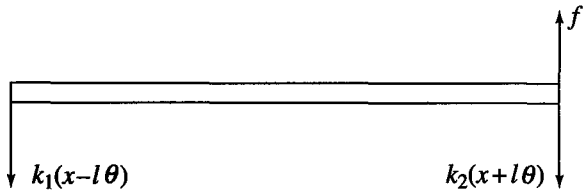


Figure 4.9 Free body diagram for an autonomous vehicle

give us (Figure 4.9)

$$m\ddot{x} = -k_1(x - l\theta) - k_2(x + l\theta) + f$$

and

$$\bar{I}\ddot{\theta} = -k_2(x + l\theta)l + k_1(x - l\theta)l + fl$$

Rearranging this yields:

$$m\ddot{x} + k_1(x - l\theta) + k_2(x + l\theta) = f$$
$$\bar{I}\ddot{\theta} + k_2(x + l\theta)l - k_1(x - l\theta)l = fl$$

or

$$m\ddot{x} + (k_1 + k_2)x + (k_2 - k_1)l\theta = f \tag{4.3.10}$$
$$\bar{I}\ddot{\theta} + (k_1 + k_2)l^2\theta + (k_2 - k_1)lx = fl \tag{4.3.11}$$

Although with more complicated systems we'd have to find our solutions numerically, we can get a closed-form solution when we've only got two equations to deal with. Assuming solutions of $x(t) = \bar{x}\cos(\omega t)$, $\theta(t) = \bar{\theta}\cos(\omega t)$ to go along with the forcing $f = \bar{f}\cos(\omega t)$, (4.3.10) and (4.3.11) will become

$$(k_1 + k_2 - \omega^2 m)\bar{x} - +(k_2 - k_1)l\bar{\theta} = \bar{f} \tag{4.3.12}$$

and

$$(k_2 - k_1)l\bar{x} + (k_1 l^2 + k_2 l^2 - \omega^2 \bar{I})\bar{\theta} = \bar{f}l \tag{4.3.13}$$

Solving (4.3.13) for $\bar{x}$ in terms of $\bar{\theta}$ gives

$$\bar{x} = \frac{\bar{f}l - (k_1 l^2 + k_2 l^2 - \bar{I}\omega^2)\bar{\theta}}{(k_2 - k_1)l} \tag{4.3.14}$$

Substituting (4.3.14) into (4.3.12) and solving for $\bar{\theta}$ yields

$$\bar{\theta} = \frac{\bar{f}l(2k_1 - m\omega^2)}{\Delta(\omega)} \tag{4.3.15}$$

where

$$\Delta(\omega) = (k_1 + k_2 - m\omega^2)(k_1 l^2 + k_2 l^2 - \bar{I}\omega^2) - l^2(k_2 - k_1)^2$$

Using (4.3.15) in (4.3.14) gives us

$$\bar{x} = \frac{\bar{f}(2k_1l^2 - \bar{I}\omega^2)}{\Delta(\omega)} \tag{4.3.16}$$

Now we're in a position to do some design work. We'll assume that the instruments are αl meters to the right of the bar's center. Thus the displacement is given by $y = x + \alpha l\theta$ for small angles of oscillation, where α runs from -1 to 1. We can use this information, together with (4.3.15) and (4.3.16), to express the displacement of the instruments as

$$y = \bar{f}\left(\frac{2k_1l^2(1+\alpha) - (\bar{I} + \alpha ml^2)\omega^2}{\Delta(\omega)}\right) \tag{4.3.17}$$

Our minimization problem is now pretty straightforward. Once we're given the frequency of motor vibration, we can use (4.3.17) to determine the placement at which y will be zero. And $y = 0$ in (4.3.17) means that the numerator must equal zero. Thus we'll have

$$2k_1l^2(1+\alpha) - (\bar{I} + \alpha ml^2)\omega^2 = 0$$

or, solving for α in terms of ω,

$$\alpha = \frac{\bar{I}\omega^2 - 2k_1l^2}{2k_1l^2 - \omega^2ml^2}$$

If we're told that the forcing frequency is 55 rad/s, then the optimal placement of the instrument package will be $\alpha = .105\,l$, or .105 m to the right of center.

Example 4.4

Problem In this example we'll consider another kind of problem. The system will be identical to that of Example 4.3. The difference in this case is that instead of a force applied, we'll now have a sensor mounted. The purpose of the sensor will be to tell us what's happening in the overall system. For specificity, we'll assume that the sensor is an accelerometer that reads vertical accelerations. The question before us is where *not* to place the sensor.

Solution The way to approach this problem is to ask what the worst situation might be. We know that the system will support two modes of vibration. If we want complete knowledge of what's going on, the sensor must be able to respond to both these modes. Therefore the worst placement of the sensor would be those places on the bar for which no information is transmitted for one or both modes.

Rewriting (4.3.10) and (4.3.11) in matrix form, we have

$$\begin{bmatrix} m & 0 \\ 0 & \bar{I} \end{bmatrix}\begin{Bmatrix} \ddot{x} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & (k_2 - k_1)l \\ (k_2 - k_1)l & (k_1 + k_2)l^2 \end{bmatrix}\begin{Bmatrix} x \\ \theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Thus our eigenvector equations are

$$\begin{bmatrix} k_1 + k_2 - m\omega^2 & (k_2 - k_1)l \\ (k_2 - k_1)l & (k_1 + k_2)l^2 - \bar{I}\omega^2 \end{bmatrix}\begin{Bmatrix} \bar{x} \\ \bar{\theta} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Solving for the eigenvectors associated with the two natural frequencies (using the parameter values of Example 4.3) yields

$$\omega_1 = 30.7 \text{ rad/s}, \quad X_1 = \begin{Bmatrix} .9600 \\ .2799 \end{Bmatrix}, \quad \omega_2 = 55.3 \text{ rad/s}, \quad X_2 = \begin{Bmatrix} -.0967 \\ .9953 \end{Bmatrix}$$

Now we can ask where on the beam the deflection is equal to zero for each of these modes. The general displacement along the beam is the same as it was for Example 4.3, $x + \alpha l\theta$. For the first eigenvector this means that the displacement is proportional to $.9600 + .2799\alpha$, where α measures displacement to the right of the midpoint in meters. The displacement is going to be equal to zero when

$$\alpha = -\frac{.9600}{.2799} = -3.430$$

Is this a ridiculous answer? After all, it says that the zero point occurs off the bar, way to the left. Well, the answer is that it's not ridiculous at all; what it's telling us is that there aren't any points on the bar that don't move when the bar is oscillating in the system's first mode. Our accelerometer can be placed anywhere and still pick up the motion from the first mode. The situation, however, is different for the second mode. For this mode, the displacement is proportional to $-.0967 + .9953\alpha$. This is zero when $\alpha = \frac{.0967}{.9953} = .0972$. Thus there is a node (point of zero motion) .0972 m to the right of center. If we place our accelerometer there, then there is no chance for it to pick up the second mode's motion.

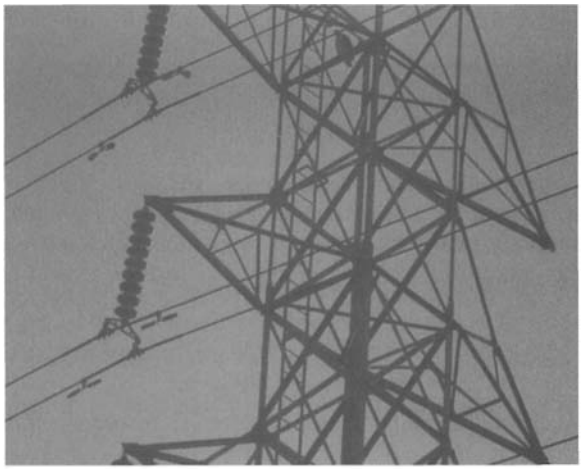
Our conclusion is that we should put the accelerometer anywhere but .0972 m to the right of center. Of course, physical considerations tell us that the displacement of the second mode is pretty small in that point's surrounding neighborhood as well. Thus we'd want to position the accelerometer a reasonable distance away.

The effect we've just seen is a very important one in vibrational analyses. For complicated systems, we'll have a large number of locations for which the response of a particular mode will be very small. Knowing where these vibrational nodes occur is crucial to place our sensors in effective locations. In the same way, if you want to control a particular mode then you wouldn't want to apply the force to a node either, since this would keep the force from influencing that mode.

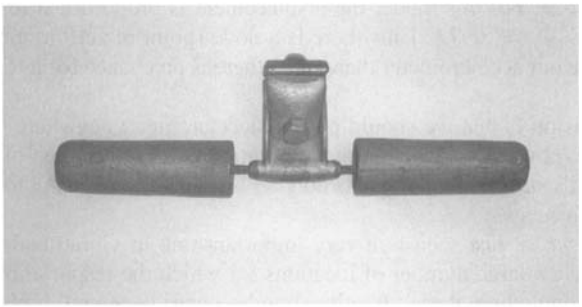
4.4 VIBRATION ABSORBERS WITHOUT DAMPING

We've already seen how altering the mass, spring, and damping properties of an SDOF system will alter its response. This section will discuss an interesting twist in which we control the response of a sinusoidally forced mass by adding more spring and mass elements to the system. We won't need to add damping, and we won't be dissipating more energy to control the response. Instead, we'll distribute the vibrational energy, moving it from the original mass to the new one.

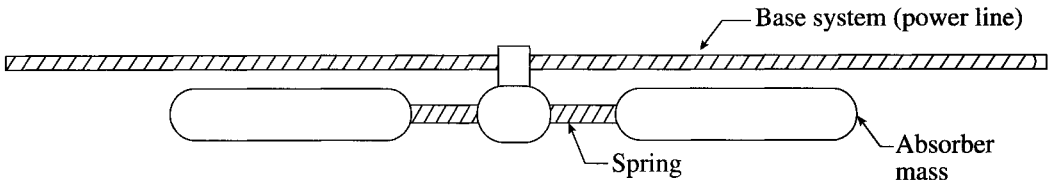
Devices that absorb energy in this way are found in a variety of applications. One common use is to reduce the vibrational response of high tension power lines. Figure 4.10*a* shows a photo of a typical powerline with attached vibrational absorbers, known as Stockbridge dampers. Figure 4.10*b* shows a close-up of the device, and Figure 4.10*c* shows a simplified diagram of the total powerline/vibration absorber system. Keep in mind that, as already mentioned, there is no energy dissipation taking place,



(*a*)



(*b*)



(*c*)

Figure 4.10 (*a*) Stockbridge dampers an powerlines, (*b*) Close-up of a stockbridge damper and (*c*) Elements comprising a stockbridge damper

in spite of the name. All that is happening is that energy which *was* vibrating the cable, will now vibrate the Stockbridge damper.

A simplified picture of a vibration absorber application is shown in Figures 4.11. Figure 4.11*a* indicates the original system, in which a spring-mass combination (m_1 and k_1) is acted upon by a

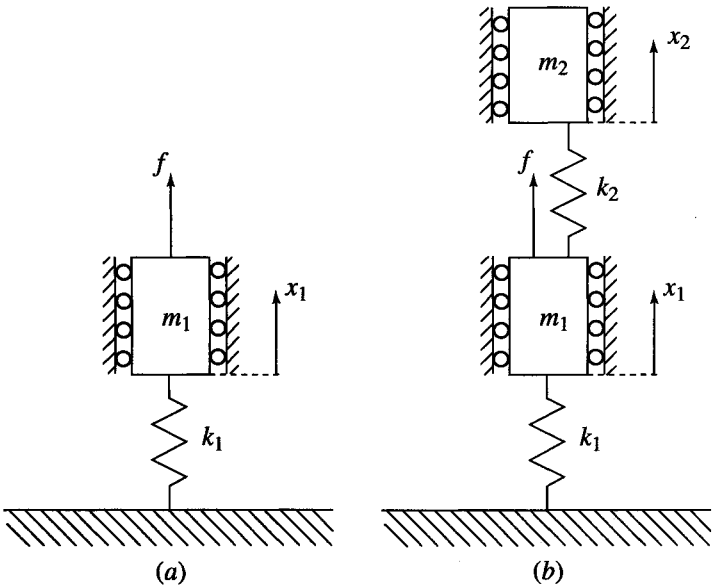


Figure 4.11 (*a*) Primary S-M system; (*b*) undamped vibration absorber, primary and secondary S-M systems

sinusoidal force. The vibration absorber consists of the mass m_2 and the spring k_2, with the total system illustrated in Figure 4.11*b*. We can picture the original system as representing any one of a variety of situations. For the purposes of illustration, let's presume that it represents an engine on its mount, and the sinusoidal force is the force due to a rotating imbalance in the engine. The equation of motion for the original system is given by

$$m_1\ddot{x} + k_1 x = \bar{f}\sin(\omega t) \tag{4.4.1}$$

and the solution is

$$x(t) = \frac{\bar{f}}{k_1 - m_1\omega^2}\sin(\omega t) \tag{4.4.2}$$

To be even more specific, let's assign values to m_1, k_1, $\bar{f}$, and ω. We'll let $m_1 = 100$ kg, $k_1 = 900$ kg/s^2, $\bar{f} = 1$ N, and $\omega = 3$ rad/s. What's the result of these choices? Well, since we've chosen the driving frequency to be equal to the system's natural frequency, we're in a resonance condition and the response becomes unbounded. Figure 4.12 shows the displacement response as a function of frequency. As you can see, the response goes to infinity at $\omega = 3$ rad/s.

This state of affairs is clearly *not* good. It is, however, a reasonable situation to consider. Even if the engine doesn't usually operate at the natural frequency of the engine/base combination, it may happen in some applications that this frequency must be passed through as the engine comes up to speed. Possibly this is also the frequency at which the engine operates most efficiently. Of course, one solution to the problem might be to redesign the suspension. Unfortunately, the suspension design might well be constrained by other design considerations. Thus, what we'll do instead is investigate how we can mitigate the vibration problem by adding a subsystem (m_2 and k_2) to the engine. This

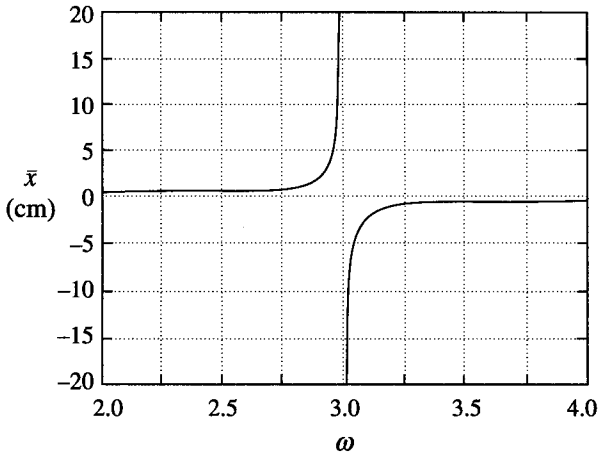


Figure 4.12 Response of primary mass before the addition of vibration absorber

subsystem will be attached to m_1 in the manner shown in Figure 4.11*b*. What we now have is a 2 DOF problem rather than the original SDOF one. The m_1/k_1 system is referred to as the primary system, and the m_2/k_2 system is known as the secondary system.

Since this problem is simply a special case of the more general problem worked out in the Section 4.3, we can substitute our particular values for m_1, k_1, etc. and obtain the solutions

$$\bar{x}_1 = \frac{(k_2 - m_2\omega^2)\bar{f}_1}{\Delta_1(\omega)} \tag{4.4.3}$$

and

$$\bar{x}_2 = \frac{k_2\bar{f}_1}{\Delta_1(\omega)} \tag{4.4.4}$$

with $\Delta_1(\omega) \equiv (k_1 + k_2 - m_1\omega^2)(k_2 - m_2\omega^2) - k_2^2$.

We can glean a wealth of observations from this particular case. First, we see from (4.4.3) that the response, $\bar{x}_1$, can be *zero* if the forcing frequency, ω, is equal to $\sqrt{\frac{k_2}{m_2}}$. Since we have complete freedom over how we choose m_2 and k_2, we can easily choose them such that $\frac{k_2}{m_2} = \frac{k_1}{m_1}$. Think of what this implies. It means that if our system used to be in a resonant condition because the driving frequency was equal to the SDOF's natural frequency $\left(\frac{k_1}{m_1}\right)$, the newly created 2 DOF system will actually have a zero amplitude response at that frequency. Thus, instead of reducing the amplitude from infinity to something finite, as the addition of damping would have done, we've actually reduced it to zero. Although it isn't obvious right now, adding a degree of freedom as we've done will always create one natural frequency above the original natural frequency and one below it. The frequency at which the zero amplitude response occurs will always lie between the two and thus can always be placed where the original natural frequency occurred.

Before moving beyond this point, we should think physically about what's going on here. If the amplitude $\bar{x}_1$ is zero, that means that m_1 is completely stationary. So we can view our system

as not being the forced 2 DOF problem that it really is, but as the SDOF problem of a mass, connected by a spring to a stationary mass. This is shown in Figure 4.13. In this view, the forcing has disappeared (since it is applied to the stationary mass) and all we're left with is a free vibration problem.

We can also see why the mass m_1 remains stationary. Although it isn't *actually* fixed in place, we're forcing it at $\omega = \sqrt{\frac{k_1}{m_1}}$. As we've mentioned, this would be the natural frequency of oscillation of m_2 and k_2 if they had been attached to an immovable body. Thus the m_2/k_2 combination oscillates at just the right frequency to cancel out the external force, and this is exactly what happens. The phase of the absorber force is exactly opposed to that of the external forcing. By oscillating at the correct amplitude, the magnitude of the forcing is also matched. Thus m_1 experiences two forces acting on it that are exactly opposed to each other. The end result—no motion. These forces are illustrated in Figure 4.14.

Let's calculate this out just to be sure. The external force applied to m_1 is $\bar{f}\cos(\omega_n t)$, where $\omega_n^2 = \frac{k_1}{m_1} = \frac{k_2}{m_2}$. From (4.4.4) we have $\bar{x}_2 = \frac{k_2 \bar{f}_1}{\Delta_1(\omega)}$. Since m_1 is stationary, the force acting on m_1 due to k_2 is equal to $k_2 x_2$. Using $\omega = \omega_n$ to evaluate $\Delta_1(\omega)$ gives us $-\bar{f}\cos(\omega_n t)$ as the force being exerted on m_1 by k_2. Thus the forces due to the external forcing and the moving mass m_2 are exactly equal and opposed, resulting in no overall unbalanced force acting on m_1.

We can gain some additional insight by looking at two particular absorbers. Figure 4.15 shows the response of both the primary and absorber masses for the case $k_2 = 90$ N/m and $m_2 = 10$ kg. What this means is that we've changed the mass of the overall system by 10%. As you can see, $\bar{x}_1$ goes

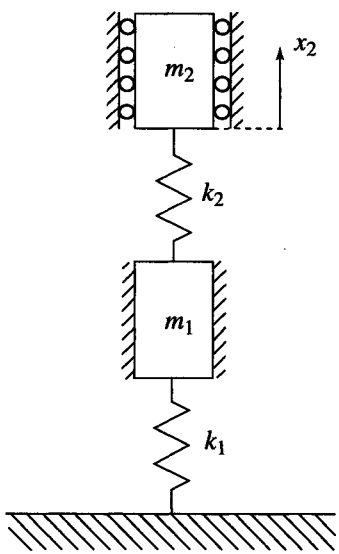


Figure 4.13 Physical equivalent to the effect of an absorber spring-mass

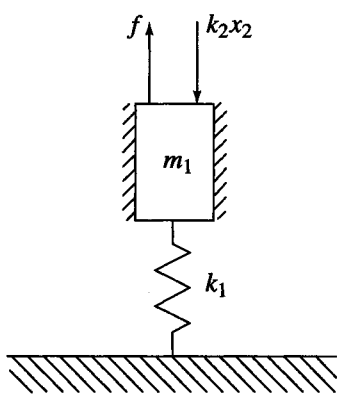


Figure 4.14 Opposed forces acting on primary mass

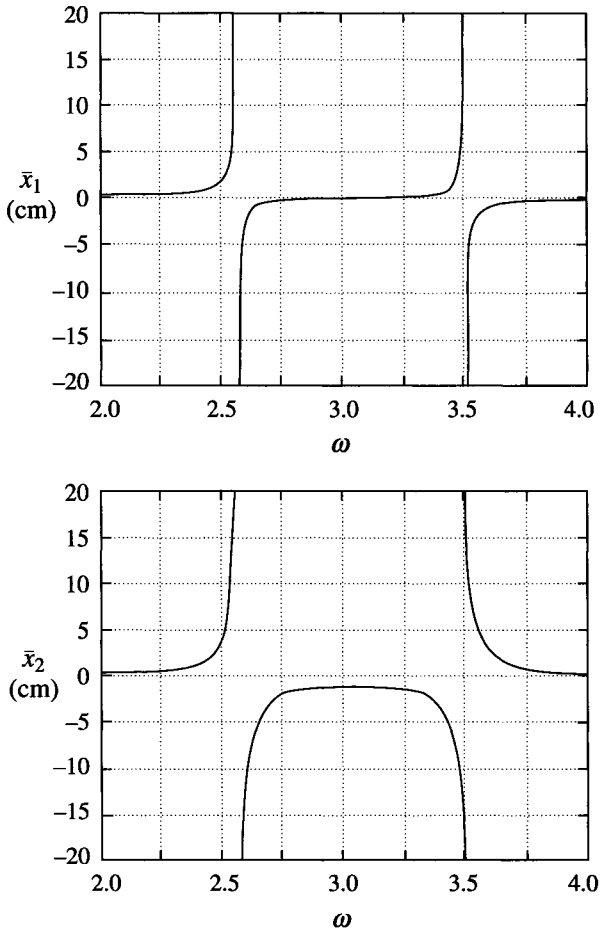


Figure 4.15 Response of vibration absorber with absorber mass equal to one-tenth of primary mass

through zero at $\omega = 3$ rad/s, as predicted. Furthermore, there is a reasonably broad band of frequencies about 3 rad/s for which the response level is quite low. Thus, even in the face of fluctuations in the exciting frequency, we'll still have only a small response. Figure 4.15 also shows the response of the absorber mass itself ($\bar{x}_2$). Again, as the equations predict, there is no frequency for which the amplitude is zero.

Although these results look pretty good, you might be wondering if we could do even better. After all, the only requirement for a vibration absorber to work is that $\sqrt{\frac{k_2}{m_2}}$ equal 3 rad/s. Although the 10% weight penalty we just used isn't terribly bad, wouldn't it be better to use even less mass? After all, designers are often very concerned about weight, and if the actual mass doesn't matter for a vibration absorber, then why not make it as small as possible.

Figure 4.16 shows what happens when we reduce both k_2 and m_2 by a factor of 10. In this case, the secondary mass is a hundred times smaller than the primary mass, meaning that we've hardly

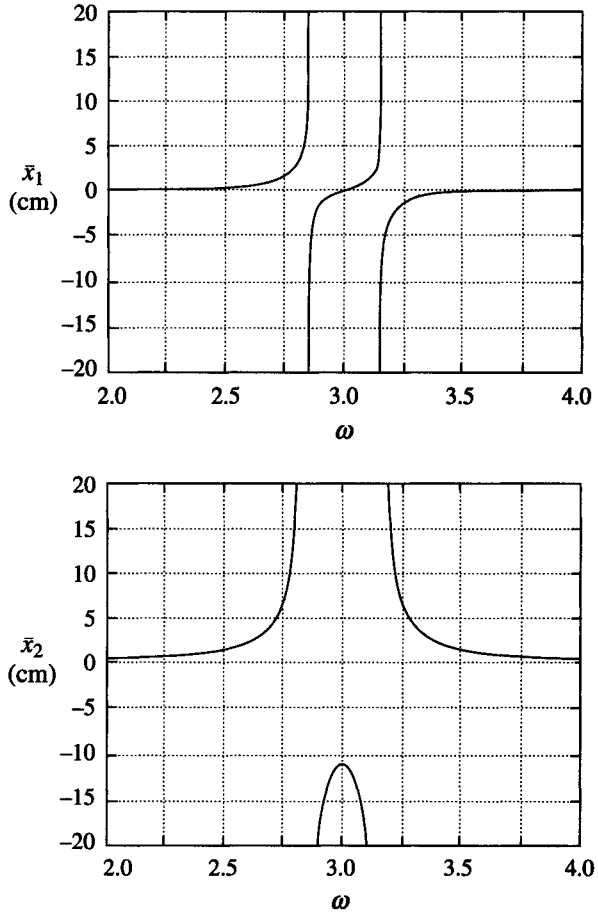


Figure 4.16 Response of vibration absorber with absorber mass equal to one one-hundredth of primary mass

changed the weight of the system at all. But as Figure 4.16 shows, we have to pay a price for the weight savings. First off, the two resonant frequencies are closer together than they were for the larger absorber. Instead of a relatively wide range of frequencies for which the absorber is effective, we now have a very tight range. Only very close to 3 rad/s do we have small vibrations for our primary mass. If the excitation frequency were to shift a bit, we could again end up with a vibration problem, since we'd then encounter resonance conditions around $\omega = 2.7$ rad/s and $\omega = 3.2$ rad/s.

The other downside is seen in the plot of $\bar{x}_2$ vs ω. By decreasing the mass and stiffness, we've increased the amplitude of $\bar{x}_2$ by a factor of 10 at $\omega = 3$ rad/s. This is, in fact, a direct consequence of our having reduced the spring stiffness: k_2 is the means through which the vibrational force is generated that counters the applied forcing. Since the force generated is equal to the stiffness times the amplitude of the motion, if you lower the stiffness by a factor of 10, then to compensate, you've got to increase the amplitude by the same factor.

These results show us that the familiar problem of trading off advantages and problems exists for vibration absorbers. To create a wide band of absorption, you need a heavy mass and a stiff spring. Along with the wide absorption band, you also have a small secondary mass motion. Therefore the packaging concerns aren't very significant but the weight concerns are. If you use a small mass and spring then you'll have a narrow absorption band, meaning that the absorber isn't terribly "robust" to changes in the parameters of the system. Furthermore, packaging will now be a problem because of the large deflections that the secondary mass will experience. As a designer, you'll have to balance these competing trends to come up with an acceptable overall design.

Example 4.5

Problem Consider again the two-mass system of Figure 4.4. No force acts on m_1 and a force $f_2 = \bar{f}\cos(\omega t)$ acts on m_2. $k_2 = 10,000$ N/m, $k_3 = 0$, $m_1 = 20$ kg, and $m_2 = 10$ kg. What must k_2 be so that m_2 will be stationary when $\omega = 60$ rad/s?

Solution For m_2 to be stationary, the applied force must be counteracting the free vibrations of the constrained system, as now illustrated.

The natural frequency of the constrained system is given by

$$\omega_n = \sqrt{\frac{k_1 + k_2}{m_1}}$$

Since ω must equal ω_n in a vibration absorber problem, we have

$$\omega = 60 = \sqrt{\frac{k_1 + k_2}{m_1}} = \sqrt{\frac{10,000 + k_1}{20}}$$

$$3600 = \frac{10,000 + k_1}{20}$$

$$k_1 = 62,000 \text{ N/m}$$

4.5 REAL BEHAVIOR OF A VIBRATION ABSORBER

We've just seen that vibrations can be avoided in a forced mass by the addition of a vibration absorber. However you might be a bit doubtful. After all, the second mass could counter the externally applied force only because it was oscillating exactly out of phase with this force. The obvious question is then, What if the force is *not* exactly out of phase with the second mass? In this case, we'll have an unbalanced force on the first mass, and it should move. How does the real world get around this problem?

Figure 4.17 shows the actual response (found from numerically integrating the equation of motion) of both the first and second masses of a vibration absorber problem for a system shown earlier (Figure 4.11*b*). Note that the primary mass does not remain stationary. However, it's important to realize that the vibrations are *bounded*. If the force and the second mass weren't correctly phase-aligned, we'd have a resonance condition and the oscillations would grow. Since we certainly don't

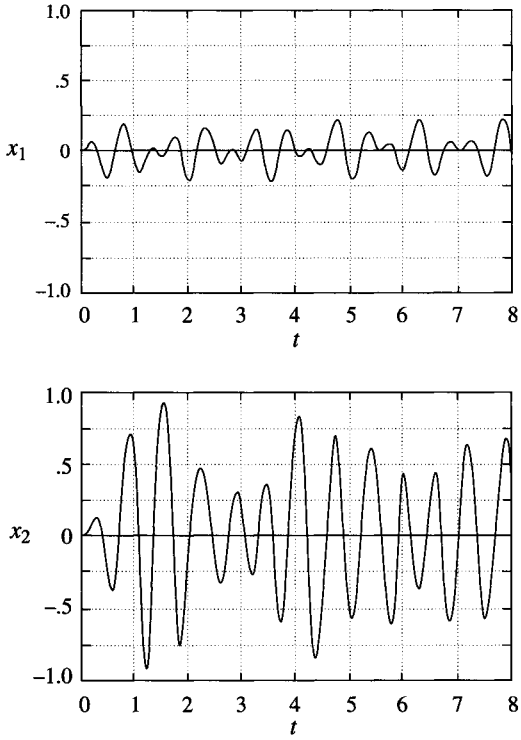


Figure 4.17 Actual response of undamped vibration absorber

have growing oscillations, the force and the second mass *are* correctly phase-aligned. The vibrations occur because with no damping in the system, the homogeneous solution (that due to initial conditions) never dies out; it "rings" forever. This is the vibration we're seeing—that due to initial conditions. To avoid this solution, we can carefully choose our initial conditions to obtain the vibration absorption we want. If the initial conditions are chosen so that the spring force is exactly opposing the applied force, then we'll see purely sinusoidal oscillations for x_2 and no motion for x_1.

Since we can't precisely set up the initial conditions we might need for a real-world implementation, must we assume that vibration absorbers don't actually work, that we'll always have transient vibrations due to initial conditions? Luckily, we don't have to assume this at all, and the reason was just alluded to. We'll run into this problem only in the absence of damping. Since purely undamped systems *never* exist in the real world, we should really add some damping to the system and see how it behaves. Figure 4.18 shows the same system as before with a small amount of linear damping added to the first mass. A typical response of this system is shown in Figure 4.19. Quite a different result! Now we see that after a transient phase, the system responds just as predicted, with a stationary first mass and a second mass that oscillates continually to cancel the external forcing, just what we'd have seen if the conditions had been perfectly set up.

There are only two problems left that we need to address. First, since we've changed the system by adding damping, how do we know that the system is behaving according to prediction? We predicted

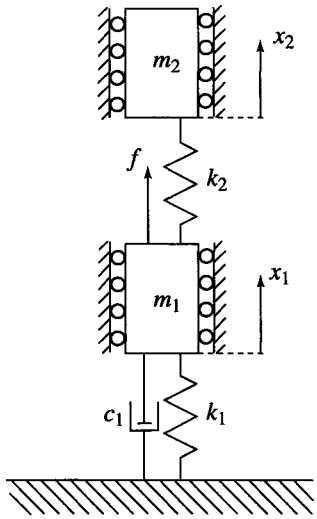


Figure 4.18 Vibration absorber with one damper

only that the *undamped* system, not a damped one, would act in the way shown. Second, just why is it that damping allowed the system to give us the response we predicted? The first question is most easily answered. Remember our physical discussion of the absorber problem: if the first mass is kept stationary, then we're left with an eigenvalue problem for the second mass. That's how we determined that the correct spring/mass ratio for the absorber should be one for which the natural frequency was equal to the forcing frequency of the external force. Luckily, since the damper is attached between the first mass and ground, it doesn't move if the first mass is stationary. So we've got the *same* eigenvalue problem as before. The damper is operative only if the first mass is moving; once it stops, the damper drops out of the picture. As you can see from looking at Figure 4.19, the first mass is certainly moving at the start of the numerical simulation. Thus the damper is operative and must in fact be responsible for bringing about the vibration cancellation that is going on later in the simulation. We're therefore stuck now with determining why this happens. The nice point about this second question is that it will shed a great deal of light on the action of damping in vibratory systems and will serve as a good motivation for the damping analyses to come in later sections.

The best way to approach this second question is to focus our attention on the *desired solution* (stationary first mass) and ask how perturbations $\Delta x_i(t)$ away from this solution will behave. Thus we'll assume that our solution looks like

$$x_1(t) = 0 + \Delta x_1(t) \tag{4.5.1}$$

$$x_2(t) = a\cos(\omega_2 t) + \Delta x_2(t) \tag{4.5.2}$$

for a forcing

$$f(t) = -ak_2\cos(\omega_2 t) \tag{4.5.3}$$

where $\omega_2 = \sqrt{\frac{k_2}{m_2}}$.

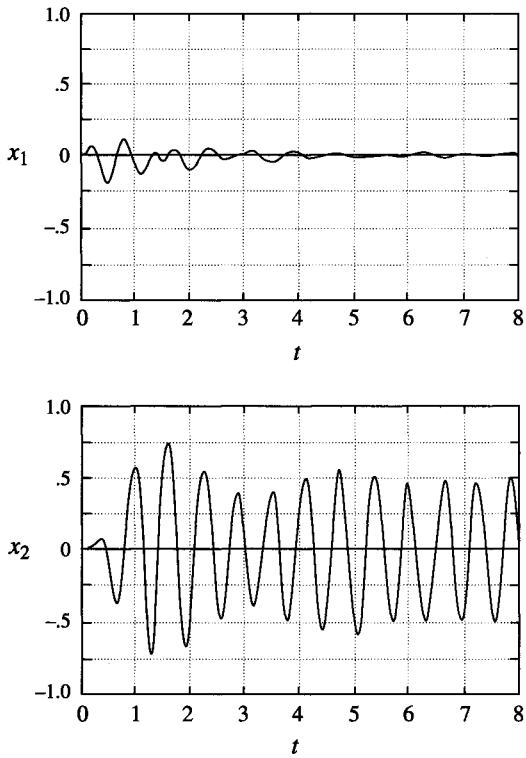


Figure 4.19 Actual response of damped vibration absorber

The approach we're using is a very important one. Since we're basing our assumed solution on a known, *steady state* solution, the added terms (Δx_1 and Δx_2) are the motions that occur *in addition* to the steady state solution, i.e., the transient behavior. Put another way, we want to capture the behavior of the system from $t = 0$ to some future time t_1. We already know that a steady state solution exists for which the first mass is stationary and the second acts to cancel the applied external force. We also know that the actual solution will have a transient part and a steady state part. Since we already know what the steady state part is, we can explicitly include it in our assumed solution, leaving the unknown Δx_i's to represent the transient solutions. This approach to finding the transient behavior of a system is quite common in nonlinear analyses and will be of equal use here.

Our governing equations of motion are

$$m_1\ddot{x}_1 + c_1\dot{x}_1 + (k_1 + k_2)x_1 - k_2x_2 = f(t) \tag{4.5.4}$$

and

$$m_2\ddot{x}_2 + k_2x_2 - k_2x_1 = 0 \tag{4.5.5}$$

Substituting our assumed solutions into (4.5.4) and (4.5.5), we find

$$m_1\Delta\ddot{x}_1 + c_1\Delta\dot{x}_1 + (k_1 + k_2)\Delta x_1 - k_2(a\cos(\omega_2 t) + \Delta x_2) = -ak_2\cos(\omega_2 t) \tag{4.5.6}$$

and

$$m_2(-a\omega_2^2\cos(\omega_2 t) + \Delta\ddot{x}_2) + k_2(a\cos(\omega_2 t) + \Delta x_2) - k_2\Delta x_1 = 0 \tag{4.5.7}$$

You can see that all the steady state solutions, along with the forcing, cancel out of the equations. (For instance, in (4.5.6) the $-k_2 a\cos(\omega_2 t)$ term on the left cancels out the $-ak_2\cos(\omega_2 t)$ term on the right.) Thus, upon removing these known terms, we're left with equations that simply involve the Δx_i's:

$$m_1\Delta\ddot{x}_1 + c_1\Delta\dot{x}_1 + (k_1 + k_2)\Delta x_1 - k_2\Delta x_2 = 0 \tag{4.5.8}$$

and

$$m_2\Delta\ddot{x}_2 + k_2\Delta x_2 - k_2\Delta x_1 = 0 \tag{4.5.9}$$

Now, instead of the forced problem we had to deal with before, we've got a free vibration problem. If you examine these equations, you'll see that they're exactly the same equations you would have derived if you'd been concerned with free vibrations of Figure 4.11*b* in the absence of any external forcing. Thus we've made the very important observation that the transient motion in a linear, forced system is governed by the *free vibrations* of that system when no forcing is present.

We've already seen in our work with SDOF systems that damping alters the pure sinusoids that occur for undamped problems into damped sinusoids. The same thing occurs with MDOF systems. Rather than begin a detailed look at the general effect of damping in a system, we're going to postpone this topic until we've looked at normal forms in the next section . Thus we'll content ourselves with looking at the numerical response of (4.5.8) and (4.5.9). Figure 4.20 shows the response of these equations to an arbitrarily chosen set of initial conditions. As you can see, the responses simply oscillate and decay. Eventually, both Δx_1 and Δx_2 approach zero. What does this mean for our vibration problem? It means that as t becomes large, the transient solutions $\Delta x_1(t)$ and $\Delta x_2(t)$ die away, leaving the steady state solution in which the first mass is stationary, i.e., the desired solution for a vibration absorber. It's important to realize that there were no approximations involved in this analysis. Thus it is *always* true that the transient solution will die down as governed by (4.5.8) and (4.5.9), leaving the desired steady state solution.

Let's recap a bit to realize what we've gotten into. First, we've seen that damping, which always exists in the real world, can be quite helpful, not just because it allows disturbances to a system to die down, but also because it can be the mechanism through which passive vibrational control can be effected. In the case just examined, the vibrational response of the system with damping was the same as that predicted for the system without damping. We should now be motivated to ask if this is always true (it's not) and to try and understand the different ways in which damping affects the response of MDOF systems. This is (almost) the next item we'll be discussing. However, before trying to jump into this topic, it will be helpful to look more closely at two more items. First we'll generalize our vibration absorber observations to the multidimensional case. Then we'll look at the different ways of expressing general solutions, specifically, to see if we might be able to view MDOF systems in a

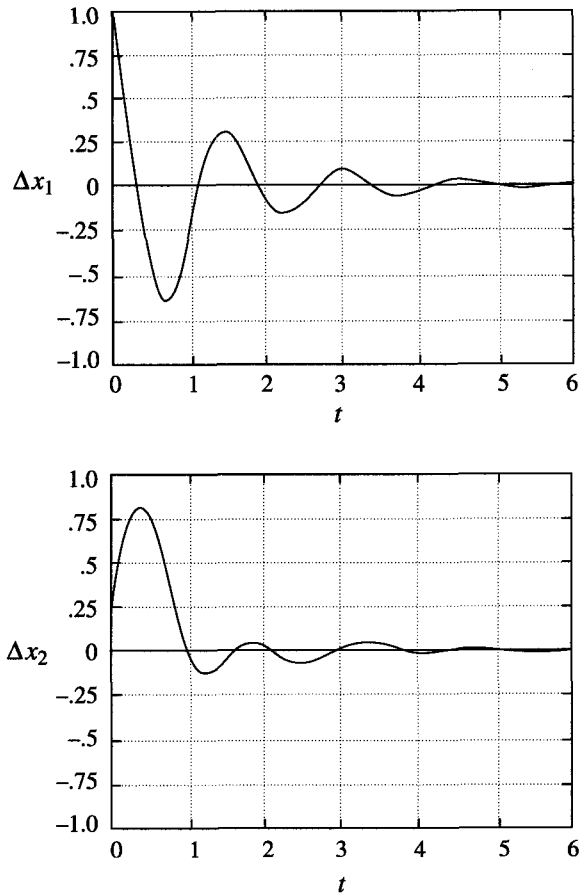


Figure 4.20 Transient solution behavior of a damped vibration absorber

simpler way, i.e., as sets of completely independent, single-degree-of-freedom systems. After we've tackled these issues, we'll return to the question of damping.

4.6 ZEROS IN A FORCED RESPONSE

In Section 4.4 we determined what frequencies of excitation will cause the response at the driven mass to equal zero for a 2 DOF system. The problem was stated differently, of course. We said that our original SDOF system had severe vibrations, and we determined how to reduce the vibrations to zero by the addition of a second mass-spring assembly. The end result, however, is the same as if we'd started with a 2 DOF system and asked what frequency of forcing, applied to the first mass, would result in that mass being stationary. You'll recall that in the process of solving the problem we noticed that, as far as the system response was concerned, a stationary mass was the same as a rigidly fixed mass. Thus the correct frequency of excitation turned out to be the natural frequency of the remaining mass and springs.

The same thing can be seen in MDOF systems. In an MDOF system, we have n masses. Our question will become, What frequency of force excitation, applied to a particular mass, will cause a particular mass in the system to have no motion? Consider the serial chain of mass and springs shown in Figure 4.21a. A sinusoidal force is applied to m_1, and we want to determine the frequencies of excitation at which m_1 will be stationary. Then we'll ask whether m_2 can also be stationary. After we've examined the different masses, we'll move the force and again see how the individual masses might be stationary, even though a sinusoidally varying force is being applied to the system.

Our first task is to see for what frequencies m_1 remains unmoving. Just as in the vibration absorber, we can realize that if m_1 is still, it's as if k_2 is attached to a rigid wall at the left in Figure 4.21b. We're left with a three-mass system: $m_2 - m_4$ with $k_2 - k_5$. Since this is a 3 DOF system, it will have three natural frequencies and three eigenvectors. This is the answer we're seeking. m_1 can be stationary if the applied force exactly opposes the forces that k_2 will transmit (which are due to the free vibrations of the three free masses).

Note that when we had two masses, we were able to get the response of one of the masses to be zero at only one frequency. Furthermore, we saw that the response of the second mass was never zero if the force was applied to the first. In this case we've seen that there are four masses in the system, and we can cause the forced mass to have zero response at three frequencies. For both cases, the forced mass can be zero at $n - 1$ frequencies, where there are n masses in the system.

What if we apply the force to the second mass, as shown in Figure 4.22a rather than to the first?, Are there $n - 1$ frequencies at which the second mass has zero response? If the second mass is stationary, then we've got an SDOF system on the left (made up of k_1, k_2, and m_1) and a 2 DOF system on the right (made up of k_3, k_4, k_5, m_3, and m_4) (Figure 4.22b). Thus we'll have one natural frequency from the left and two from the right. Again we've got $n - 1$ frequencies for which we get a zero amplitude response. This finding is a general one. When we have n masses, the number of frequencies at which the driven mass is stationary is $n - 1$.

What about if we move off from the driven mass? For instance, let's consider the situation of Figure 4.21a, where the force is applied to m_1. This time, let's suppose that m_2 is stationary. Since m_2 isn't moving (by assumption), we have a 2 DOF system to the right (m_3 and m_4) that can oscillate at one of its two natural frequencies as in Figure 4.21c. This vibration will cause a force on m_2 due to k_3. The only way to keep m_2 from moving in response to this force is to apply an equal and opposite force from the left. This force can easily be supplied by appropriately forcing m_1. Thus we have 2, or $n - 2$, zeros for m_2 when m_1 is forced. We saw the same thing for the 2 DOF vibration absorber. When the first mass was forced, the second mass had no zeros responses. For that case n was equal to 2 and therefore $n - 2 = 0$, telling us that there are no frequencies at which the second mass would be stationary.

If you look at m_3 instead of m_2 you'll see that the number of frequencies that yield a zero responses is equal to $n - 3$, i.e., there will be just one (Figure 4.21d). Finally, when you force the first mass and look at the mass furthest away (m_4 in this case) there will be no frequencies at which the response is zero.

All these deductions can be seen in Figure 4.23. This figure shows the actual response versus forcing frequency for each mass of the system of Figure 4.21a. For these plots all the m_i and k_i were

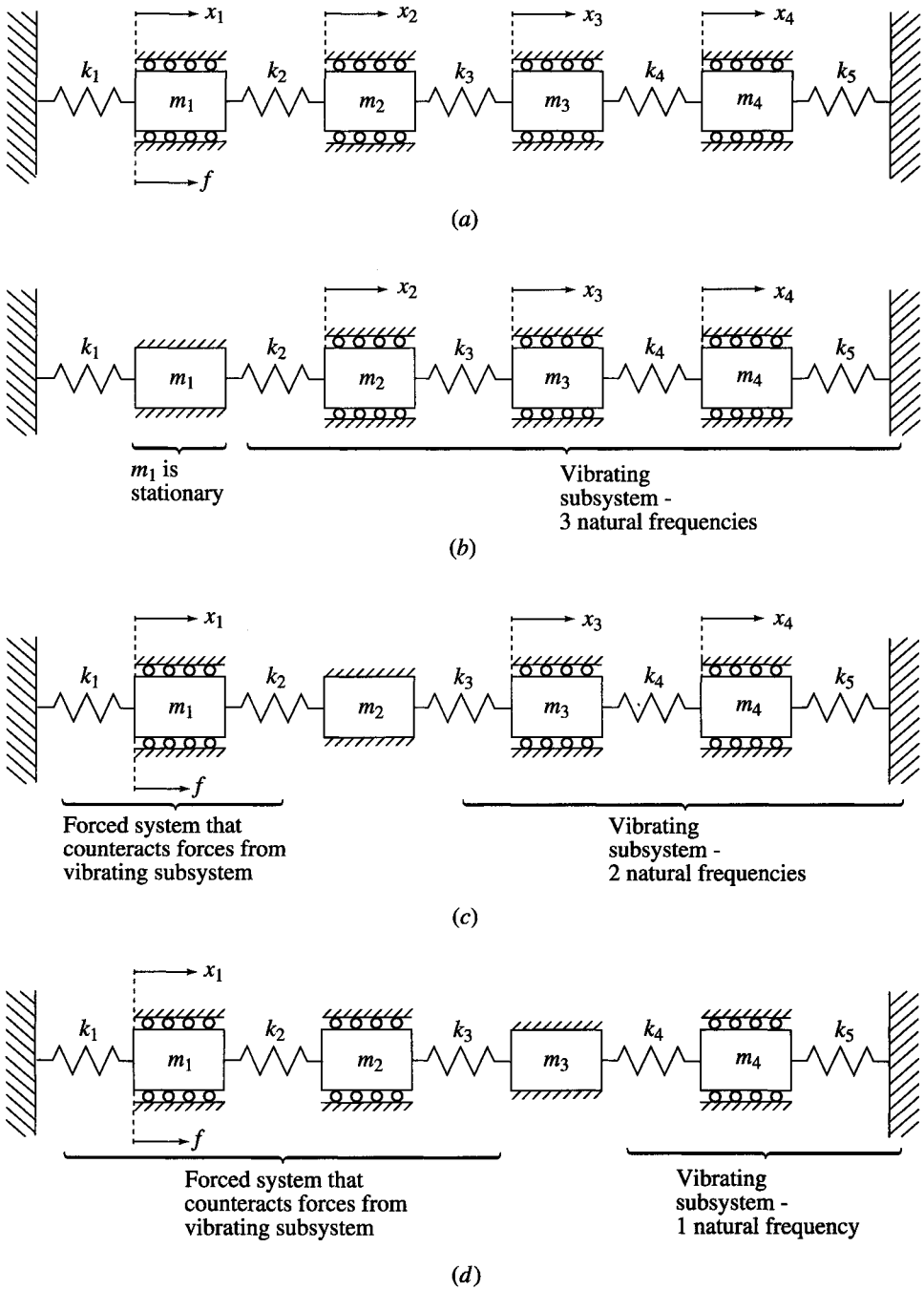


Figure 4.21 Serial chain of spring-mass elements: (*a*) Force applied to first mass and (*b*) m_1 stationary, force acting on m_1 (*c*) m_2 stationary, force acting on m_1 and (*d*) m_3 stationary, force acting on m_1

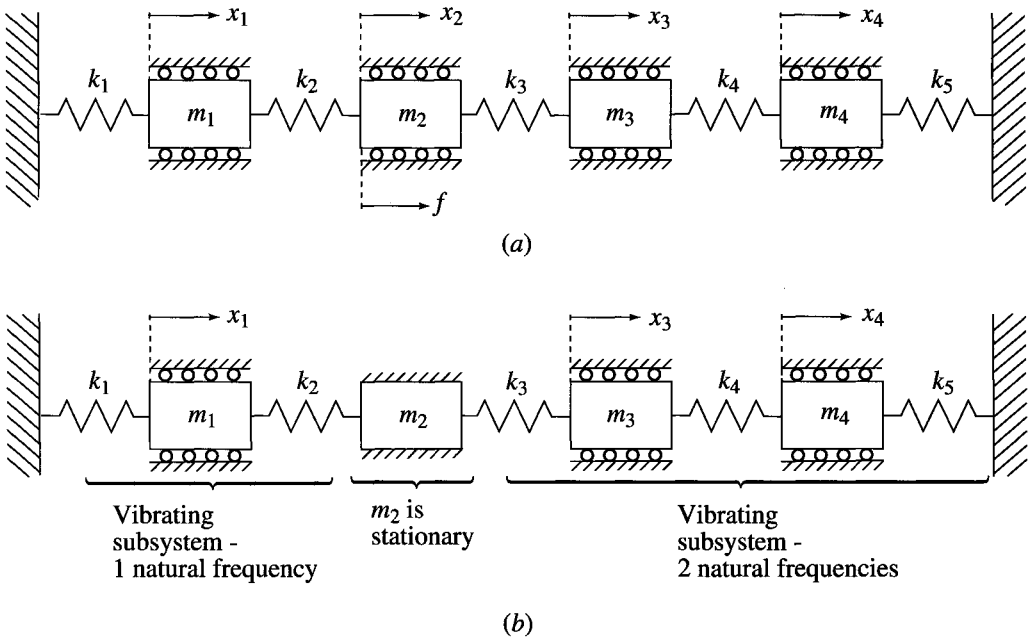


Figure 4.22 Serial chain of spring-mass elements: (*a*) force applied to second mass, (*b*) m_2 stationary, force acting on m_2

given unit values and the magnitude of f was also set equal to 1.0. Figure 4.23*a* shows the response of m_1 for a force applied to m_1. You can see that the response is equal to zero at three distinct frequencies. Moving from m_1 to m_2, Figure 4.23*b* loses one of the zeros. Instead of going through zero at around $\omega = 1.4$, the response approaches zero, but then turns around and moves away from zero and out to negative infinity. Figure 4.23*c* displays only a single zero. Note that this zero occurs at around $\omega = 1.4$. Thus we see that the zeros don't drop off in a regular fashion as we move away from the forced mass: m_2 (in Figure 4.23*b*) didn't have a zero at the same place that m_3 (Figure 4.23*c*) *does* have one. But m_3 has lost the zeros where m_2 had them. Lastly, Figure 4.23*d* displays no zeros.

These observations give us some insight into the design of our systems. We might well want a zero response somewhere in the system. Perhaps we need to mount a mirror and don't want any vibrations that would disturb it. Or it may be that a sensitive piece of equipment will be mounted in a particular place and we want to minimize any vibrations. The foregoing analysis will tell us how many frequencies there are that will permit a vibration absorption to take place.

If we call the forced mass our *actuated* mass and the mass whose output we're observing our *sensed* mass then what we've got is a sensor/actuator pair. By having the sensor and actuator be the same mass, we've created what is called a *collocated sensor/actuator pair.* The ability to be collocated is highly desirable in any control system, and the reason is pretty clear. What we normally want to do in a control system is keep some part of the system under control, usually implying that the part under control is kept at zero, even in the face of disturbances. Figure 4.23*d* shows the worst case of

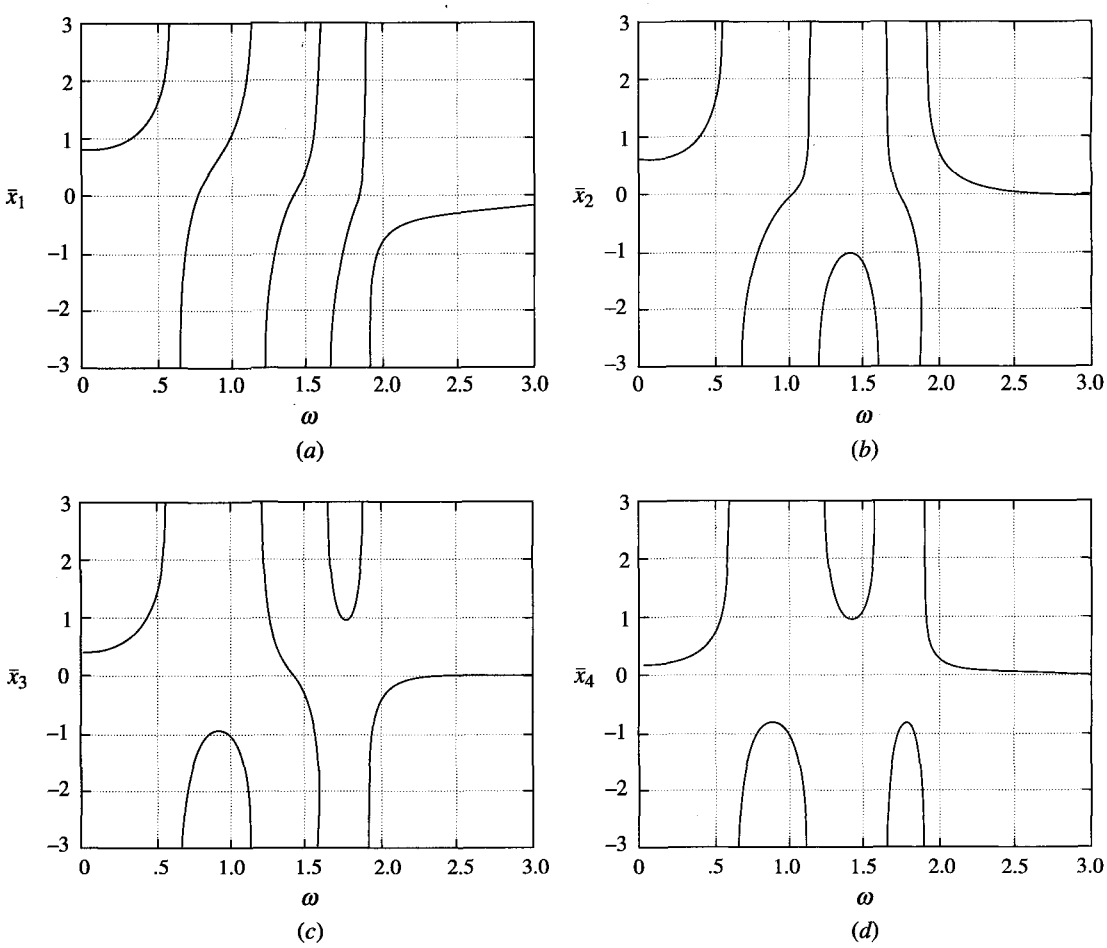


Figure 4.23 Four graphs showing amplitude responses for different masses in a four-mass system

noncollocation. In this situation the control force and sensing are located as far from each other as possible; they're separated by the dynamics of the entire system. This makes control very difficult, as reflected by the absence of any frequencies at which the response is zero.

The physical analogue of this situation is a golfer or a fly fisherman. In each case the person is holding the device (be it a flexible golf club or a really flexible fishing rod), and the part to be controlled is at the other end of the device (the club head or the end of the rod). We could model either device as a string of masses and springs and, if we assume that the person is watching the tip with a view to controlling the system, we'd have a completely noncollocated setup. (Interesting side note: You'll no doubt be saying—if you're a golfer or a fisherman—that people don't actually observe the tip of the implement when golfing or fishing. You're right. The reason is that the dynamics and control problems are too complex. They're beyond our abilities to control in any sort of feedback

sense. That's why they're done in what's called an *open-loop* fashion. We simply learn by rote what the correct motions are and then repeat them mechanically.)

When running a modal test we'll also have a similar situation in that we have an exciter and sensor that we use to gather the modal data. We'll have either a collocated or a noncollocated setup depending upon whether sensor and exciter act on the same mass. Precise collocation usually is difficult to achieve, although a new type of sensor, a *self-sensing* actuator, has recently been developed and manages to achieve true collocation.

Example 4.6

Problem Determine how many zeros to expect in the system shown in Figure 4.22*a* when the sensed mass is m_3. For this example $m_1 = .75$, $m_2 = .5$, $m_3 = .9$, $m_4 = .55$, $k_1 = 1.1$, $k_2 = 2$, $k_3 = 1$, $k_4 = 1$, and $k_5 = 3.3$. Masses are given in kilograms and stiffnesses in newtons per meter.

Solution First we'll look to the right of m_3. Since m_3 is stationary, by implication the k_4, k_5, m_4 single-degree-of-freedom system could be oscillating at its natural frequency. This will impress a force on m_3 by means of k_4. If k_3 is impressing an equal and opposite force (because of the force f) then m_3 can remain stationary. That's one frequency. Are there others? Let's look to the left of m_3 to find out. For there to be another frequency, we're going to have m_2 be stationary as well as m_3. If m_2 wasn't stationary, then its motion would make m_3 move, violating our stationarity assumption. There is only one frequency at which m_2 could move without moving m_3, and that is the natural frequency of the k_4-k_5-m_4 system. However we've already taken that answer into account. Thus we conclude that if there is to be a second zero for m_3, m_2 must be stationary. Is this possible? Very definitely. The force on m_2 can be exactly counteracting free vibrations of the k_1-k_2-m_1 system. Physically, the k_1-k_2-m_1 system will be oscillating and the external force will exactly cancel the forces due to this oscillation that k_2 is impressing on m_2. Thus we see that m_2 and m_3 are both stationary. So, in fact, is m_4, since there's no way for it to get excited. This immediately tells us that we've found out something else, namely, that this zero for m_3 must also be a zero of m_2 and m_4.

Figure 4.24 shows the forced response for this system. As you can see, we've got just two frequencies that correspond to zero motion (around $\omega = 2$ rad/s and $\omega = 2.75$ rad/s.) Both zero frequencies that were predicted have shown up.

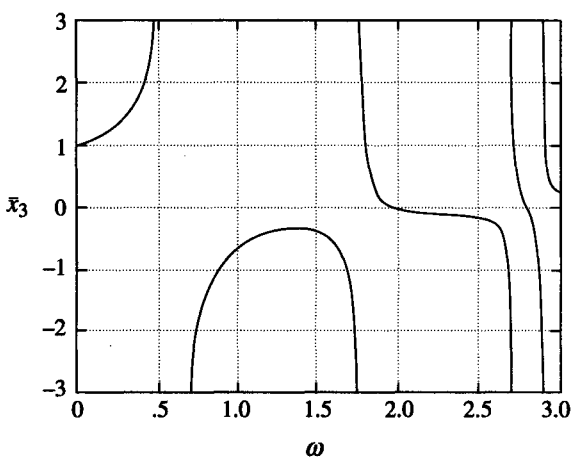


Figure 4.24 Graph showing zero amplitude for third mass, force applied to second mass

An interesting fact that can be seen in the preceding examples is that the number of zeros in a mass-spring chain is always equal to $n-(s+1)$, where n is the number of masses and s is the number of masses away the sensed mass is from the driven mass. In our first example the driven mass was m_1 and the sensed mass varied from m_1 to m_4 (Figure 4.23). When m_1 was the sensed mass, there was no separation between sensed and forced masses, therefore $s = 0$. The number of zeros was three, which matched the result predicted from evaluating $n - 1$. When the sensed mass was m_2, we had $s = 1$ and $n - (1 + s) = 2$, matching the numerical results. In Example 4.6, the separation between the driving and forced mass was again equal to one, and $n-(s+1) = 2$, agreeing with the number of zeros shown in Figure 4.24. This formula also tells us to expect no zeros when the driven and sensed masses are at opposite ends of the chain. In this case $s = n-1$ and $n-(s+1)$ is always equal to zero.

As an example of how this kind of analysis is being used [5], buildings can be analyzed with regard to active control during earthquakes by modeling the different floors as lumped masses, connected by spring/damper elements, with large mechanical actuators used to counter the effect of ground motion and render the floors stationary. The problem setup is essentially identical to the one discussed in this section.

4.7 PUTTING PROBLEMS INTO NORMAL FORM

So far, the way we've viewed 2 DOF systems has been reasonably similar to the way we've dealt with SDOFs. We've seen that 2 DOF systems support eigenvector responses, but we haven't particularly exploited this fact. Now we'll do so.

One obvious characteristic of the general 2 DOF systems we've seen so far is that the system equations are coupled. It's impossible to solve for the response of x_1 without at the same time solving for x_2 (where x_1 and x_2 are the two coordinates of our system). We even have names for the particular types of coupling we might run into. For instance, the following equation is said to be *mass coupled*, or *inertially coupled*:

$$\begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 8 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} \bar{f}_1 \\ \bar{f}_2 \end{Bmatrix} \tag{4.7.1}$$

because the two equations are coupled through the $[M]$ matrix. In a like manner, (4.7.2) is *stiffness coupled* or *spring coupled* because the coupling comes through the $[K]$ matrix.

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 7 & -2 \\ -2 & 8 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} \bar{f}_1 \\ \bar{f}_2 \end{Bmatrix} \tag{4.7.2}$$

A fully uncoupled set of equations would look like

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 8 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} \bar{f}_1 \\ \bar{f}_2 \end{Bmatrix} \tag{4.7.3}$$

For this case, we see that the equation governing the response of x_1 has nothing to do with x_2's behavior. Thus we can solve either equation independently of the other; they're uncoupled.

It's clear that uncoupled equations are nice to have, since it's always easier to solve two independent equations than it is to solve a simultaneous set of equations. The question is, Can we somehow transform our particular equations into a form in which they're decoupled? Obviously, the answer is yes or I wouldn't be mentioning it. When the equations are fully decoupled we say the system is in *normal form.* To take a specific example, let's look at the problem illustrated in Figure 4.25. We'll consider the case for which $m_1 = m_2 = 1$ and $k_1 = k_2 = k_3 = 1$. The governing equations are:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} \bar{f}_1 \\ \bar{f}_2 \end{Bmatrix} \cos(\omega t) \tag{4.7.4}$$

This set of equations is stiffness coupled. To decouple them, we're going to define a new set of coordinates. Instead of x_1 and x_2, we'll express the equations of motion in terms of η_1 and η_2, where the η_i's are defined as

$$\eta_1 = \tfrac{1}{2}(\sqrt{2}x_1 + \sqrt{2}x_2) \tag{4.7.5}$$

and

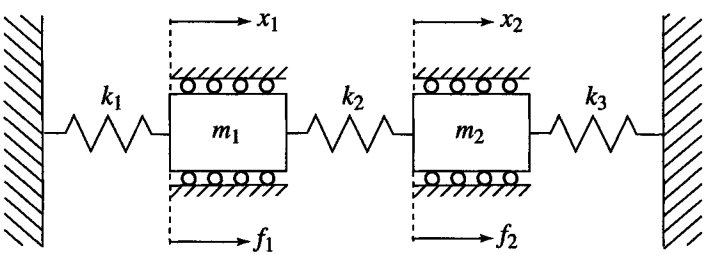
$$\eta_2 = \tfrac{1}{2}(\sqrt{2}x_1 - \sqrt{2}x_2) \tag{4.7.6}$$

This can be expressed in matrix form as $H = [V]X$, where

$$H \equiv \begin{Bmatrix} \eta_1 \\ \eta_2 \end{Bmatrix}, \quad X \equiv \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}, \quad \text{and} \quad [V] \equiv \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{4.7.7}$$

At the moment, it shouldn't be clear why in the world we'd want to do this, but we're going to anyway. Later, we'll see why this particular transformation was chosen. Before moving on from this point, we should think about what this new coordinate set is all about. η_1 is simply $\sqrt{2}$ times the mean of x_1 and x_2, and η_2 is $\sqrt{2}$ times one-half of the difference between x_1 and x_2. Thus, if $x_1 = 1$ and $x_2 = 1$, then $\eta_1 = \sqrt{2}$ and $\eta_2 = 0$. Recall that the two eigenvectors for this system are

$$X_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}, \quad X_2 = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \tag{4.7.8}$$



$m_1 = m_2 = 1$ kg
$k_1 = k_2 = k_3 = 1$ N/m

Figure 4.25 Forced, 2 DOF system

Thus, when only the first eigenvector is involved, $x_1 = x_2 = 1$ and therefore $\eta_1 = \sqrt{2}$ and $\eta_2 = 0$. Similarly, when only the second eigenvector is present, $\eta_1 = 0$ and $\eta_2 = \sqrt{2}$, since $x_1 = 1$ and $x_2 = -1$. Therefore the η coordinates are doing a great job of telling us which eigenvector is present in the response. When η_1 is nonzero and η_2 equals zero, we know that only the first eigenvector is involved. The same holds when $\eta_1 = 0$ and η_2 is nonzero, except that it's the second eigenvector that's now involved. And if both eigenvectors are present, the η's tell us how much of each is present.

This is *not* a trivial point. Previously, all we knew were the positions of m_1 and m_2. We could use this information to determine what was happening with regard to the eigenvectors of the system, but it required some calculation. Now we have an immediate reading with respect to the eigenvector's participation through the η coordinates. Of course in this case we don't know what the individual x_1 and x_2 responses are and we'd have to calculate them from our knowledge of the η's. We can get this information by multiplying by the inverse of the transformation matrix V, so that instead of $H = [V]X$ we have

$$X = [U]H \tag{4.7.9}$$

where

$$[U] = [V]^{-1} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \tag{4.7.10}$$

If we go ahead and use (4.7.9) in (4.7.4), we'll get

$$[M][U]\ddot{H} + [K][U]H = \bar{F}\cos(\omega t) \tag{4.7.11}$$

Now there's one more thing we need to do. Rather than leave our equations in the form of (4.7.10) and (4.7.11), we're going to premultiply by $[U]^T$ (the transpose of $[U]$) to obtain

$$[U]^T[M][U]\ddot{H} + [U]^T[K][U]H = [U]^T\bar{F}\cos(\omega t) \tag{4.7.12}$$

We can express this more compactly as

$$[M']\ddot{H} + [K']H = \bar{F}'\cos(\omega t) \tag{4.7.13}$$

where

$$[M'] \equiv [U]^T[M][U], \quad [K'] \equiv [U]^T[K][U], \quad \text{and} \quad \bar{F}' \equiv [U]^T\bar{F} \tag{4.7.14}$$

If you carry through with the indicated operations you'll find that (4.7.13) has the following matrix entries:

$$[M'] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad [K'] = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}, \quad \text{and} \quad \bar{F}' = \begin{bmatrix} \frac{1}{2}(\bar{f}_1' + \bar{f}_2') \\ \frac{1}{2}(\bar{f}_1' - \bar{f}_2') \end{bmatrix} \tag{4.7.15}$$

Note the good news—we've decoupled the equations. The equation for η_1 is completely independent of the equation for η_2 and vice versa. Our equations are in formal form and the η_i's are our normal

coordinates. Furthermore, we can notice some interesting facts. First the mass matrix $[M']$ is the identity matrix. This isn't a big deal, since it was already equal to the identity matrix when we started. However, what's interesting is that we can work it so that *any* initial mass matrix is transformed into the identity matrix. Obviously this property will save storage space on a computer if it's true for any size system (and it is). Second, we can see that the entries in the $[K']$ matrix are exactly equal to the squares of the natural frequencies. It seems that this coordinate transformation is quite powerful. It makes the mass matrix equal to the identity matrix, produces a diagonal stiffness matrix having the square of each natural frequency on the main diagonal and, to top it off, gives us a direct readout of the modal character of the response through the η coordinates.

Having seen all there useful features, you may well be interested in why and how they come about, and how you can harness this transformation power for yourself. Amazingly, the procedure is pretty simple. Just look at the transformation matrix $[U]$ for a second and ask yourself what's familiar about the columns of this matrix. Yes, that's right! They're simply equal to the eigenvectors of the original system. That's almost all there is to it. We find the eigenvectors, put them side by side to define $[U]$, premultiply by $[U]^T$, and get our decoupled form. There are a couple of additional fine points, which we'll talk about as we rigorously show why this decoupling takes place. If you're completely uninterested in seeing this, you can skip to Section 4.9 now. (But you'll miss some good stuff!)

Example 4.7

Problem Use the transformation matrix

$$[U] = \begin{bmatrix} .3606 & .7326 \\ .4541 & -.6785 \end{bmatrix}$$

to put the system

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 100 & -20 \\ -20 & 80 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} \cos(10t)$$

into normal form.

Solution

$$[U]^T[M][U] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$[U]^T[K][U = \begin{bmatrix} 22.95 & 0 \\ 0 & 110.4 \end{bmatrix}$$

$$[U]^T F = \begin{bmatrix} 1.27 \\ -.624 \end{bmatrix}$$

Thus we have

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{\eta}_1 \\ \ddot{\eta}_2 \end{Bmatrix} + \begin{bmatrix} 22.95 & 0 \\ 0 & 110.4 \end{bmatrix} \begin{Bmatrix} \eta_1 \\ \eta_2 \end{Bmatrix} = \begin{Bmatrix} 1.27 \\ -.624 \end{Bmatrix} \cos(10t)$$

4.8 ORTHOGONALITY OF SYSTEM EIGENVECTORS

You're probably all familiar with orthogonality as the term is applied to spatial vectors. It means that the vectors are oriented at right angles to each other. Mathematically, when two vectors are orthogonal, their dot product is zero. Additionally, one way of calculating the dot product of two column vectors A and B is by evaluating $A^T B$. These notions are very close to those that we'll develop in this section. The only twist is that instead of saying certain vectors are orthogonal just to each other, we'll see that the orthogonality conditions include the $[M]$ and $[K]$ matrices. Thus orthogonality is very tightly tied to the particular system we're examining.

We'll sneak up on these results in a very straightforward way, namely, by making a simple examination of our equations and then drawing some conclusions from them. For clarity, we'll derive our results with a 2 DOF example. Keep in mind though that the approach is applicable to a problem with any number of degrees of freedom, as we'll indicate after finishing with the 2 DOF example.

To begin, we'll look at our free vibration problem

$$[M]\ddot{X} + [K]X = 0 \tag{4.8.1}$$

We know now that the solutions to this problem define our system eigenvectors (X_1 and X_2) and natural frequencies (ω_1 and ω_2). Thus, writing (4.8.1) in terms of these known solutions gives us

$$-\omega_1^2[M]X_1 + [K]X_1 = 0 \tag{4.8.2}$$

and

$$-\omega_2^2[M]X_2 + [K]X_2 = 0 \tag{4.8.3}$$

Rearranging these equations gives us

$$\omega_1^2[M]X_1 = [K]X_1 \tag{4.8.4}$$

and

$$\omega_2^2[M]X_2 = [K]X_2 \tag{4.8.5}$$

What we've got here are two sets of vector equalities (the vector on the left equals the vector on the right). Clearly, we'll still have an equality if we premultiply (4.8.4) by X_2^T and (4.8.5) by X_1^T. It's not obvious yet *why* we'd want to do this, but there's certainly nothing that says we can't. So now we have

$$\omega_1^2 X_2^T[M]X_1 = X_2^T[K]X_1 \tag{4.8.6}$$

and

$$\omega_2^2 X_1^T[M]X_2 = X_1^T[K]X_2 \tag{4.8.7}$$

Now let's take the transpose of (4.8.7). (Yes, I know. This seems pointless. But it's not, so just hold on for another moment or two.) Transposing (4.8.7) gives us

$$\omega_2^2 X_2^T [M]^T X_1 = X_2^T [K]^T X_1 \tag{4.8.8}$$

Since the $[M]$ and $[K]$ matrices of spring-mass vibratory systems are symmetric, this is the same as

$$\omega_2^2 X_2^T [M] X_1 = X_2^T [K] X_1 \tag{4.8.9}$$

Take a second to compare (4.8.6) and (4.8.9). They are *identical* except that one involves ω_1 while the other contains ω_2. We're now in a position to be clever. Let's subtract (4.8.9) from (4.8.6). This will give us

$$(\omega_2^2 - \omega_1^2) X_2^T [M] X_1 = 0 \tag{4.8.10}$$

This is interesting! It tells us something fundamental about our eigenvectors. If $\omega_2 \neq \omega_1$, then $X_2^T [M] X_1$ must equal zero. (Otherwise (4.8.10) wouldn't be satisfied.) This is the start of our orthogonality condition. For vibrations problems, the eigenvectors aren't directly orthogonal to each other (i.e., it's not true that $X_1^T X_2 = 0$); rather, they're orthogonal with respect to the mass matrix $[M]$. But wait, there's even more. From (4.8.9) we see that if $X_2^T [M] X_1 = 0$, then $X_2^T [K] X_1$ must also equal zero. Thus the eigenvectors are orthogonal with respect to both the stiffness *and* the mass matrices. In other words, if $\omega_1 \neq \omega_2$ then $X_1^T [M] X_2 = X_1^T [K] X_2 = 0$.

The next thing we need to ask is whether the eigenvectors are also orthogonal to themselves. That is, does $X_1^T [M] X_1 = 0$? One result of linear algebra is that $X^T [A] X$ is never zero if $[A]$ is positive definite unless X is itself zero. Since $[M]$ is always a positive definite matrix, this means that $X_1^T [M] X_1 \neq 0$. The precise value of $X_1^T [M] X_1$ depends on both $[M]$ and X_1. $[M]$ is fixed but, as we've discussed, X_1 can be determined only up to a point. All multiples of our chosen X_1 are also valid eigenvectors. This actually can be turned to our advantage. Rather than always have this uncertainty about X_1, we can demand that it be chosen so that $X_1^T [M] X_1 = 1.0$. This is a type of normalization (known as *mass normalization*) and, if we choose each eigenvector such that its transpose multiplied by $[M]$ multiplied by the eigenvector equals 1.0, we've got an *orthonormal* set of vectors. The "ortho" refers to the vectors' orthogonality, and the "normal" refers to the fact that they're normalized. We'll use a tilde to indicate that the eigenvectors are normalized ($\tilde{X}_i$).

Getting the orthonormalized eigenvectors isn't too hard. Say that you've got an eigenvector X_i, which isn't mass normalized, and that

$$X_i^T [M] X_i = b_i \neq 1.0$$

The mass-normalized eigenvector will just be a scalar multiple of X_i

$$\tilde{X}_i = \alpha_i X_i$$

We know that the mass-normalized eigenvector satisfies

$$\tilde{X}_i^T [M] \tilde{X}_i = 1$$

and so

$$\alpha_i^2 X_i^T [M] X_i = 1$$
$$\alpha_i^2 b_i^2 = 1$$
$$\alpha_i = \frac{1}{\sqrt{b_i}}$$

Thus

$$\tilde{X}_i = \frac{1}{\sqrt{b_i}} X_i = \frac{1}{\sqrt{X_i^T [M] X_i}} X_i$$

And there we've got our mass normalized vector. Once we do this to all the vectors we'll have our orthonormal set of eigenvectors.

Getting back to our derivation, we next we need to ask what $\tilde{X}_1^T [K] \tilde{X}_1$ is equal to. Premultiplying (4.8.4) by $\tilde{X}_1^T$ yields

$$\omega_1^2 \tilde{X}_1^T [M] \tilde{X}_1 = \tilde{X}_1^T [K] \tilde{X}_1 \tag{4.8.11}$$

Since $\tilde{X}_1^T [M] \tilde{X}_1$ is equal to 1.0, we see that $\tilde{X}_1^T [K] \tilde{X}_1$ is equal to ω_1^2. In exactly the same manner, $\tilde{X}_2^T [K] \tilde{X}_2$ is equal to ω_2^2.

Right about now is a good time to recall that one of our goals was to determine how to get a system into a decoupled form. Well, we've done it! All we need to do is make sure our transformation matrix $[U]$ is made up of our normalized system eigenvectors, i.e.,

$$[U] = [\tilde{X}_1 \vdots \tilde{X}_2] \tag{4.8.12}$$

Using X=$[U]H$ in (4.8.1) and premultiplying by $[U]^T$ gives us

$$[U]^T [M][U] \ddot{H} + [U]^T [K][U] H = 0 \tag{4.8.13}$$

which, in expanded form, is equal to

$$\begin{bmatrix} \tilde{X}_1^T [M] \tilde{X}_1 & \tilde{X}_1^T [M] \tilde{X}_2 \\ \tilde{X}_2^T [M] \tilde{X}_1 & \tilde{X}_2^T [M] \tilde{X}_2 \end{bmatrix} \begin{Bmatrix} \ddot{\eta}_1 \\ \ddot{\eta}_2 \end{Bmatrix} + \begin{bmatrix} \tilde{X}_1^T [K] \tilde{X}_1 & \tilde{X}_1^T [K] \tilde{X}_2 \\ \tilde{X}_2^T [K] \tilde{X}_1 & \tilde{X}_2^T [K] \tilde{X}_2 \end{bmatrix} \begin{Bmatrix} \eta_1 \\ \eta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4.8.14}$$

or, in terms of what we now know, these products are equal to

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{\eta}_1 \\ \ddot{\eta}_2 \end{Bmatrix} + \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{bmatrix} \begin{Bmatrix} \eta_1 \\ \eta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4.8.15}$$

Although we derived these results using a 2 DOF model, it should be clear that the results are general and can be applied to any size system. Thus, for an n DOF system, our orthogonality conditions would read

$$X_i^T [M] X_j = X_i^T [K] X_j = 0, \quad i \neq j \tag{4.8.16}$$

In this case, the matrices $[M]$ and $[K]$ are $n \times n$, X_i is the ith eigenvector, and X_j is the jth eigenvector. If the eigenvectors are mass normalized then

$$\tilde{X}_i^T [M] \tilde{X}_i = 1.0 \tag{4.8.17}$$

and

$$\tilde{X}_i^T [K] \tilde{X}_i = \omega_i^2 \tag{4.8.18}$$

We've now learned that the eigenvectors of our system are not in general orthogonal to each other, but rather are orthogonal with respect to the $[M]$ and $[K]$ matrices. To distinguish between this situation and the more common concept of orthogonal vectors, we'll make some definitions. If we've got a set of vectors that are orthogonal with respect to the $[M]$ matrix, we'll call them *mass orthogonal*. A set of vectors that are orthogonal to the $[K]$ matrix are *stiffness orthogonal*. If the set is simultaneously orthogonal to both the $[K]$ and $[M]$ matrices, then they are *system orthogonal*. Finally, if they are orthogonal to themselves, i.e.,

$$X_i^T X_j = 0 \tag{4.8.19}$$

then they are *self-orthogonal*. Note that if $[M]$ is equal to a multiple of the identity matrix, then the eigenvectors are self-orthogonal, since with $[M] = \alpha[I]$ we have

$$X_i^T [M] X_j = \alpha X_i^T [I] X_j = \alpha X_i^T X_j = 0 \tag{4.8.20}$$

This is a good time to make an observation. The eigenvectors of our system are orthogonal to the $[M]$ and $[K]$ matrices. As you might suspect, this implies that they are also independent. In fact, taken together, the eigenvectors form a very special set. If our system is n-dimensional (n physical coordinates needed to describe the system's configuration), then we can obtain *any* possible configuration by suitably combining the system eigenvectors. A simple example would be to look again at the 2 DOF system shown in Figure 4.1. If all the spring constants are identical and the masses are equal, then the eigenvectors will be $X_1 = \{1 \;\; 1\}^T$ and $X_2 = \{1 \;\; -1\}^T$. Any position of the two masses can be represented by these vectors. For instance, let's say the first mass has zero displacement and the second is moved to the right by 4 units. Then

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 4 \end{Bmatrix} = 2 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} - 2 \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \tag{4.8.21}$$

The same holds for any size system. A system's configuration can be represented by an appropriate combination of the eigenvectors. In this case, we say that the system eigenvectors *span* the configuration space of the system. It's worth noting that you don't need to use eigenvectors to span a system's configuration space. In fact, all you need is a set of independent vectors. Since the eigenvectors are independent, they work, but so will any number of other vector sets. In fact, this will be one of the main points when we introduce approximation techniques in Chapter 6.

We can show some more interesting results by using a little linear algebra. The first thing to realize is that our mass matrix is symmetric and positive definite. One linear algebra result tells us

that such a matrix can always be written as the product of two identical matrices:

$$[M] = [M]^{\frac{1}{2}}[M]^{\frac{1}{2}} \tag{4.8.22}$$

If we use this fact in our equations of motion, we'll have

$$[M]^{\frac{1}{2}}[M]^{\frac{1}{2}}\ddot{X} + [K]X = 0 \tag{4.8.23}$$

If we define a new set of coordinates by

$$[M]^{\frac{1}{2}}X = H \tag{4.8.24}$$

then we'll have

$$[M]^{\frac{1}{2}}\ddot{H} + [K][M]^{-\frac{1}{2}}H = 0 \tag{4.8.25}$$

Premultiplying by $[M]^{-\frac{1}{2}}$ leaves us with

$$\ddot{H} + [M]^{-\frac{1}{2}}[K][M]^{-\frac{1}{2}}H = 0 \tag{4.8.26}$$

or

$$\ddot{H} + [A]H = 0 \tag{4.8.27}$$

where

$$[A] = [M]^{-\frac{1}{2}}[K][M]^{-\frac{1}{2}}$$

This shows us that we can always transform our equations into a form for which the mass matrix is equal to the identity matrix and the stiffness matrix is symmetric. We know the transformed stiffness matrix must be symmetric because $[M]^{-\frac{1}{2}}$ and $[K]$ are symmetric, and products of symmetric matrices of the foregoing form are themselves symmetric.

Now let's see what we can do here. We know that the solutions of this problem will support n natural frequencies and eigenvectors. If we put the eigenvectors together into a transformation matrix $[U]$, as suggested earlier, then we can write

$$-[U][\Omega^2] + [A][U] = 0 \tag{4.8.28}$$

where $[\Omega^2]$ is an $n \times n$ diagonal matrix having the squares of the system natural frequencies on the diagonal. $[U]$ is called the *modal matrix* for the system. Pulling the left-hand term onto the other side of the equality gives us

$$[U][\Omega^2] = [A][U] \tag{4.8.29}$$

and premultiplying by $[U]^{-1}$ gives us

$$[\Omega^2] = [U]^{-1}[A][U] \tag{4.8.30}$$

What we'll do now is take the transpose of (4.8.29) and show that there is a very special relationship between $[U]^T$ and $[U]^{-1}$. Taking the transpose of (4.8.29) gives us

$$[\Omega^2]^T[U]^T = [U]^T[A]^T \tag{4.8.31}$$

But remember that both $[\Omega^2]$ and $[A]$ are symmetric. Thus (4.8.31) is equivalent to

$$[\Omega^2][U]^T = [U]^T[A] \tag{4.8.32}$$

Postmultiplying (4.8.32) by $([U]^T)^{-1}$ gives us

$$[\Omega^2] = [U]^T[A]([U]^T)^{-1} \tag{4.8.33}$$

Now compare (4.8.33) and (4.8.30). What we have are essentially two expressions of a similarity transformation, between $[A]$ and $[\Omega^2]$. Since transformations like these are unique, we see by comparing terms that $[U]^T$ must be equal to a scalar constant times $[U]^{-1}$. If we use

$$[U]^T = \alpha[U]^{-1} \tag{4.8.34}$$

in (4.8.33) we'll obtain

$$[\Omega^2] = \alpha[U]^{-1}[A][U]\frac{1}{\alpha} \tag{4.8.35}$$

or

$$[\Omega^2] = [U]^{-1}[A][U] \tag{4.8.36}$$

Note that this is exactly the same as (4.8.30). So we've found an important fact, namely, that the transpose of the modal matrix is, within a scalar multiple, equal to the inverse of the modal matrix (when the associated mass matrix is an identity matrix). Knowing this certainly allows us to save time, since calculating a transpose is substantially easier than calculating an inverse.

Example 4.8

Problem Find the modal matrix for the system

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{Bmatrix} + \begin{bmatrix} 2000 & -500 & 0 \\ -500 & 3000 & -200 \\ 0 & -200 & 2000 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

Show that $[U]^T = \alpha[U]^{-1}$.

Solution Using [U, d] = `eig`(K, M) in MATLAB produces the result

$$U = \begin{bmatrix} .3712 & .8511 & -.3714 \\ -.9166 & .3997 & 0 \\ .1485 & .3404 & .9285 \end{bmatrix}$$

Finding U^T (by typing U′) yields

$$U^T = \begin{bmatrix} .3712 & -.9166 & .1485 \\ .8511 & .3997 & .3404 \\ -.3714 & 0 & .9285 \end{bmatrix}$$

while U^{-1} (from typing `inv(U)`) produces

$$U^{-1} = \begin{bmatrix} .3712 & -.9166 & .1485 \\ .8511 & .3997 & .3404 \\ -.3714 & 0 & .9285 \end{bmatrix}$$

Since for this case $U^T = U^{-1}$, we have $\alpha = 1$.

4.9 MORE ON NORMAL FORMS

Now that we've seen how to put our system into normal form, we can go ahead and tidy up all the remaining loose ends. To start, let's consider what to do when our problem has an external forcing. In this case, the equations of motion are given by

$$[M]\ddot{X} + [K]X = F \tag{4.9.1}$$

If we go ahead and use $X = [U]H$ to decouple this set of equations, we'll obtain

$$[M']\ddot{H} + [K']H = F' \tag{4.9.2}$$

where

$$[M'] \equiv [U]^T[M][U], \quad [K'] \equiv [U]^T[K][U], \quad \text{and} \quad [\bar{F}'] \equiv [U]^T F \tag{4.9.3}$$

This was derived a few pages back ((4.7.13) and (4.7.14)) and is simply rewritten here for convenience. If the modal matrix $[U]$ has been mass normalized then $[M']$ is equal to the identity matrix and $[K']$ is equal to $[\Omega^2]$. Thus our equations are given by

$$\ddot{H} + [\Omega^2]H = F' \tag{4.9.4}$$

This is a completely decoupled problem. In this form, our problem is exactly equivalent to n individual, second-order equations. Taking the ith one at random, we have

$$\ddot{\eta}_i + \omega_i^2\eta_i = f_i' \tag{4.9.5}$$

where f_i' is simply the ith component of the modified forcing vector. We know from earlier work that η_i represents the degree of participation of the ith eigenvector in the response. Thus we can clearly see how the external force affects the various modes of the system.

Example 4.9

Problem Consider the 2 DOF example we recently looked at (in which $m_1 = m_2 = 1$ and $k_1 = k_2 = k_3 = 1$), the equations for which are given in (4.7.4). Put the system into normal form and analyze the resulting equations.

Solution We've already seen that, if the modal matrix is given by

$$[U] = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Then

$$[M'] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad [K'] = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}, \quad \text{and } [F'] = \begin{bmatrix} \frac{1}{\sqrt{2}}(f_1 + f_2) \\ \frac{1}{\sqrt{2}}(f_1 - f_2) \end{bmatrix}$$

The equations are decoupled, in just the form we expected. Let $f_1 = \bar{f}_1 \cos(\omega t)$ and $f_2 = \bar{f}_2 \cos(\omega t)$. The two relevant equations of our normal-form representation are

$$\ddot{\eta}_1 + \eta_1 = \frac{1}{\sqrt{2}}(\bar{f}_1 + \bar{f}_2)\cos(\omega t)$$

and

$$\ddot{\eta}_2 + 3\eta_2 = \frac{1}{\sqrt{2}}(\bar{f}_1 - \bar{f}_2)\cos(\omega t)$$

The solutions to these problems are simply

$$\eta_1 = \frac{\frac{1}{\sqrt{2}}(\bar{f}_1 + \bar{f}_2)}{1 - \omega^2}\cos(\omega t)$$

and

$$\eta_2 = \frac{\frac{1}{\sqrt{2}}(\bar{f}_1 - \bar{f}_2)}{3 - \omega^2}\cos(\omega t)$$

This seems to make sense. If we realize that the first mode of the system is simply both masses moving together in unison, then the way to best excite it is to have both forces acting in unison, one on each mass. Of course, in general this doesn't happen. So what counts is how much force is being applied on average to the two masses. That's what the forcing component represents—an average force. Similarly, the second mode consists of both masses moving in an opposite direction. Thus, to best excite this mode, the forces must be opposed to each other.

Let's look at a couple of cases to make sure this is clear. If both forces are acting in the same direction and with the same magnitude, then they're aimed directly at the first mode. In this case ($\bar{f}_1 = \bar{f}_2 = \bar{f}$), our equations become

$$\ddot{\eta}_1 + \eta_1 = \sqrt{2}\bar{f}\cos(\omega t)$$

and

$$\ddot{\eta}_2 + 3\eta_2 = 0$$

Thus only the first equation is being forced; the second is completely unaffected. This means that the first mode will respond and the second mode will not.

Similarly, if the forces are of equal magnitude and opposed ($-\bar{f}_2 = \bar{f}_1 = \bar{f}$), we'll have

$$\ddot{\eta}_1 + \eta_1 = 0$$

and

$$\ddot{\eta}_2 + 3\eta_2 = \sqrt{2}\bar{f}\cos(\omega t)$$

In this case, the second mode is excited while the first is unaffected.

There are two fundamental observations we can draw from all this. The first is that it is possible to excite individual modes of a system and leave others totally unaffected. This fact is often used during modal tests of large systems, such as airliners. For instance, let's say you want to perform a modal test on a large airplane. Something this big is difficult to excite effectively. One way to get a reasonably sized modal response is to place shakers along the wings and body and then vibrate the plane in such a way as to couple into just a single vibrational mode. This will give you a large, well-defined response, just what you need for a good experimental test.

The second observation is that in general, *all* modes are excited when a system is forced. As we saw in Example 4.9, only for the special cases of $\bar{f}_1 = \bar{f}_2$ and $\bar{f}_1 = -\bar{f}_2$ did we avoid exciting one mode while getting a response from another. For any other combination of inputs, both modes will be involved in the total response. This fact is also of great use in modal testing. A *very* common way of running modal tests is to hit the structure with an instrumented hammer at a particular point. Since all modes are (usually) excited by such an input, a full range of spectral data can be acquired from this single test.

4.10 LINEAR DAMPING

We've already seen that damping isn't a trivial matter, even for the SDOF case. By approximating the actual damping by a linear damping characteristic, we kept the problem solvable and were able to draw some conclusions about how damped systems behave. And, as Section 2.6 showed, damping can often be quite helpful. So let's see how we can add damping to our repertoire.

Unfortunately, the addition of one or more additional degrees of freedom causes some real complications, even if we assume linear damping. To illustrate why this is so, look at Figure 4.26, a 2 DOF spring-mass-damper system. If the dampers are removed, we're left with our familiar spring-mass example. We know that for no damping, the system has the two eigenvectors $X_1 = \frac{1}{\sqrt{2}}\{1 \;\; 1\}^T$ and $X_2 = \frac{1}{\sqrt{2}}\{1 \;\; -1\}^T$. If the system is displaced into either of these configurations, it will oscillate forever in that mode.

It would be nice if, when damping is added, the modal character of the system were preserved. Thus, we'd expect a system that was displaced into the shape of an eigenvector, to find that it *stays* in that configuration, but with the oscillations dying down at a rate that was proportional to the damping

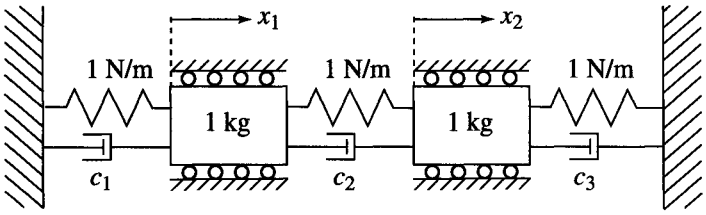


Figure 4.26 Damped 2 DOF system

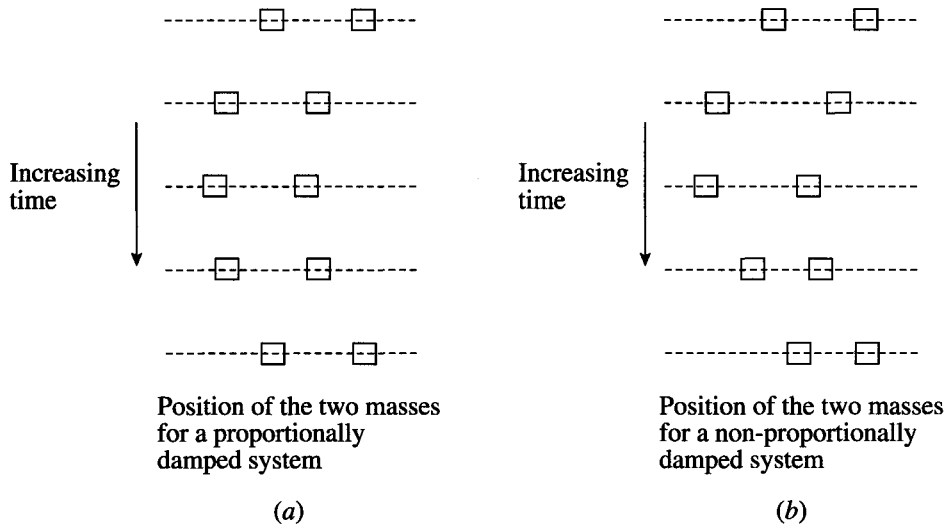


Figure 4.27 Relative displacements in nonproportionally damped system

instead of continuing forever. Unfortunately, the actual behavior is most often not like this. What more often happens can be illustrated by looking at the behavior of the system shown in Figure 4.26 with the specific damping values $c_1 = 0$, $c_2 = .2$ N·s/m, and $c_3 = .1$ N·s/m. Figure 4.27*b* shows what the resultant vibrations look like if the damped system is first displaced into the first (undamped) eigenvector's shape. As you can see, the relative displacements of the two masses are definitely *not* remaining the same: they start out equal, but later are very different.

Interestingly, although the behavior we've just seen didn't give us the modal response we were looking for, a small change in the damping matrix can remedy the situation. Instead of the damping values we used earlier, we can use $c_1 = c_2 = c_3 = .1$ N·s/m. If the two masses are equally displaced (first undamped eigenvector) and then released, we'll see what we had originally hoped for (Figure 4.27*a*). The relative displacements of the masses will always be equal as the oscillation dies down.

What this means is that some damping matrices preserve the undamped modal character of the solution and others don't. The obvious step is then to ask when we can expect the solutions to behave one way or the other.

To make this determination, let's consider the n DOF problem shown in Figure 4.28. The equations of motion for this particular system are

$$[M]\ddot{X} + [C]\dot{X} + [K]X = F \tag{4.10.1}$$

where

$$[M] \equiv \begin{bmatrix} m_1 & 0 & \cdots & 0 \\ 0 & m_2 & \cdots & \vdots \\ \vdots & \cdots & m_{n-1} & 0 \\ 0 & \cdots & 0 & m_n \end{bmatrix} \tag{4.10.2}$$

$$[C] \equiv \begin{bmatrix} c_1 + c_2 & -c_2 & 0 & 0 & \cdots & 0 \\ -c_2 & c_2 + c_3 & -c_3 & 0 & & \vdots \\ 0 & -c_3 & & & & \vdots \\ \vdots & & & & -c_{n-1} & 0 \\ & & & -c_{n-1} & c_{n-1} + c_n & -c_n \\ 0 & \cdots & & 0 & -c_n & c_n + c_{n+1} \end{bmatrix} \tag{4.10.3}$$

$$[K] \equiv \begin{bmatrix} k_1 + k_2 & -k_2 & 0 & 0 & \cdots & 0 \\ -k_2 & k_2 + k_3 & -k_3 & 0 & & \vdots \\ 0 & -k_3 & & & & \vdots \\ \vdots & & & & -k_{n-1} & 0 \\ & & & -k_{n-1} & k_{n-1} + k_n & -k_n \\ 0 & \cdots & & 0 & -k_n & k_n + k_{n+1} \end{bmatrix} \tag{4.10.4}$$

and

$$F \equiv \begin{Bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{Bmatrix} \tag{4.10.5}$$

Although the system shown in Figure 4.28 has the particular $[M]$, $[C]$, and $[K]$ entries listed earlier, (valid for a serial chain of masses), the analysis we'll be undertaking will be valid for any other configuration of masses we might wish to examine. To simplify our analysis, we'll restrict ourselves to a 2 DOF example. As usual when examining the fundamental responses of a system, we'll consider the unforced case. To illustrate the effect of damping, we'll keep the $[M]$ and $[K]$ matrices fixed and vary $[C]$.

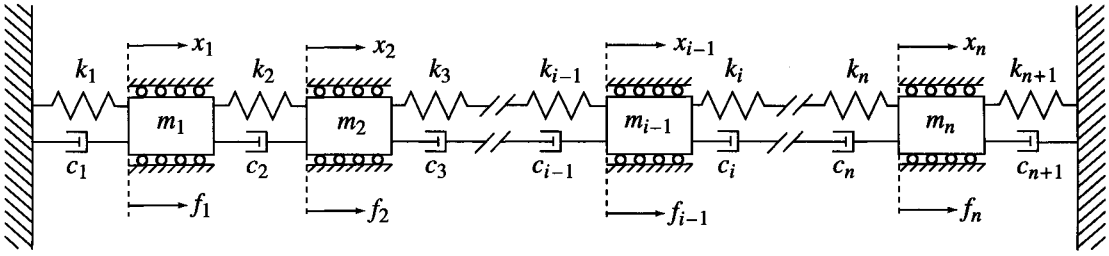


Figure 4.28 Forced n-DOF, spring-mass-damper system

The first issue that we must address is the question of how we'll be solving our problem. The equations of motion will be in the form

$$[M]\ddot{X} + [C]\dot{X} + [K]X = 0 \tag{4.10.6}$$

We could certainly attempt to solve this by assuming $X = \bar{X}e^{\lambda t}$ to obtain:

$$\left[\lambda^2[M] + \lambda[C] + [K]\right]\bar{X} = \{0\} \tag{4.10.7}$$

Unfortunately, solving this is not particularly easy. For instance, assume that we've got a 2 DOF system. With no damping, we'd have to solve a fourth order characteristic equation, *but* λ would appear only in powers of 2 (λ^2, λ^4, etc.). Thus we'd really have a quadratic in λ^2, not a fourth-order problem in λ. But now, with the addition of damping, when we take the determinant of $\lambda^2[M] + \lambda[C] + [K]$, we'll get an honest-to-goodness fourth-order equation. In a like manner, any n DOF problem will give us a $2n$th-order polynomial. And to make matters even more interesting, this polynomial will have complex solutions. This comes about because of the damping matrix's contribution ($\lambda[C]$). Thus we'd need to assume a complex solution ($\lambda = \lambda_r + i\lambda_i$) from the outset. We'd have to substitute this into our characteristic polynomial, set it equal to zero, separate all the real parts of the equation, and set them equal to zero while simultaneously setting the imaginary parts equal to zero as well. We'd then need to create a root-finder program that would solve these equations for us.

Clearly, this isn't overly easy, and it would be far preferable to use a prepackaged code. Unfortunately, there is no code that currently accepts problems in the form we're using (separate $[M]$, $[C]$, and $[K]$ matrices). Most often, a canned numerical code will want to see an eigenvalue problem in the form

$$\dot{X} = [A]X \tag{4.10.8}$$

or perhaps

$$[A]\dot{X} = [B]X \tag{4.10.9}$$

Certainly this is the way MATLAB expects to see the problem. The basic difference is that our equations are in second-order form (two differentiations with respect to time) as opposed to first order (one differentiation with respect to time). Thus, to use MATLAB or similar software, we need to get our equations into first-order form.

Those of you who have already taken a controls course will find this very familiar. The standard controls formulation is one that looks like

$$\dot{X} = [A]X + [B]U \tag{4.10.10}$$

This can be contrasted with our familiar forced representation:

$$[M]\ddot{X} + [K]X = F \tag{4.10.11}$$

The common theme behind our solution procedure for both these problems is to use linear algebra. Both our representation and the controls representation are cast as problems involving combinations of matrices and vectors, something that defines a linear algebra problem. The one advantage the controls community has over the vibrations folks is that they've *always* used the (4.10.10) formulation. We had the advantage as long as we were dealing with undamped problems, but now we're going to have to use the same form as the controls people if we want to get anywhere quickly. In fact, the MATLAB program was originally seized upon by the controls researchers as an excellent tool with which to solve their problems, and many applications have been written specifically for controls applications. Thus, all the development work has been done for us, and all we need to do is put our equations in a form MATLAB recognizes. When equations are in a form that looks like

$$[A]\dot{X} = [B]X \tag{4.10.12}$$

we say they are in *first-order form.*

One nice formulation that takes advantage of symmetry is to put our vibration problem into the form

$$[B]\dot{Y} = [A]Y \tag{4.10.13}$$

where

$$[B] \equiv \begin{bmatrix} -[K] & [0] \\ [0] & [M] \end{bmatrix}, \quad [A] \equiv \begin{bmatrix} [0] & -[K] \\ -[K] & -[C] \end{bmatrix}, \quad \text{and} \quad Y^T \equiv \{x_1 \;\; x_2 \;\; \dot{x}_1 \;\; \dot{x}_2\} \tag{4.10.14}$$

Another convention that's often used for systems that don't have too many degrees of freedom is the *state space form.* As an example of this representation, assume you're given the system equations

$$\ddot{x}_1 + 2\dot{x}_1 + 10x_1 - 5x_2 = 0$$

and

$$\ddot{x}_2 + 4\dot{x}_2 + 30x_2 - 5x_1 = 0$$

To put these equations in state space form you'd let $y_1 = x_1$, $y_2 = \dot{x}_1$, $y_3 = x_2$, and $y_4 = \dot{x}_2$. Two of the equations come from the definition of the y_i

$$\dot{y}_1 = y_2 \;\text{ and }\; \dot{y}_3 = y_4$$

The other two come from the system's equations of motion. The total representation is given by

$$\begin{Bmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \dot{y}_3 \\ \dot{y}_4 \end{Bmatrix} = \begin{Bmatrix} 0 & 1 & 0 & 0 \\ -10 & -2 & 5 & 0 \\ 0 & 0 & 0 & 1 \\ 5 & 0 & -30 & -4 \end{Bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{Bmatrix}$$

You're certainly free to choose a different representation if you wish. The main thing is to represent an n DOF vibration problem as $2n$ first-order differential equations. The next section will look more precisely at what sort of solution behavior we can expect.

Example 4.10

Problem Reexpress the following equations of motion in symmetric form and in standard state space form. Analyze them both with MATLAB and show that the final results are the same.

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 1000 & -400 \\ -400 & 2000 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Solution First we'll look at a state space form. Our state vector is given by

$$Y = \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{Bmatrix}$$

where $y_1 = x_1$, $y_2 = \dot{x}_1$, $y_3 = x_2$, and $y_4 = \dot{x}_2$.

Our equations of motion are

$$\begin{aligned} \dot{y}_1 &= \dot{x}_1 = y_2 \\ \dot{y}_2 &= \ddot{x}_1 = \tfrac{1}{2}(-1000x_1 + 400x_2) = -500y_1 + 200y_3 \\ \dot{y}_3 &= \dot{x}_2 = y_4 \\ \dot{y}_4 &= \ddot{x}_2 = 400x_1 - 2000x_2 = 400y_1 - 2000y_3 \end{aligned} \tag{4.10.15}$$

or

$$\dot{Y} = [A]Y$$

where

$$[A] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -500 & 0 & 200 & 0 \\ 0 & 0 & 0 & 1 \\ 400 & 0 & -2000 & 0 \end{bmatrix}$$

Typing [v, d] = `eig`(A) (where A = [A]) yields

$$v = \begin{bmatrix} -.0028 & -.0028 & .0457 & .0457 \\ -.1278i & .1278i & .9673i & -.9673i \\ .0219 & .0219 & .0118 & .0118 \\ .9916i & -.9916i & .2494i & -.2494i \end{bmatrix}$$

$$d = \begin{bmatrix} 45.2942i & 0 & 0 & 0 \\ 00 & -45.2942i & 0 & 0 \\ 0 & 0 & 21.1764i & 0 \\ 0 & 0 & 0 & -21.1764i \end{bmatrix}$$

We've got natural frequencies of 21.1764 and 45.2942 rad/s.

Next we'll use the symmetric formulation. Let

$$Z = \begin{Bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{Bmatrix}$$

where $z_1 = x_1$, $z_2 = x_2$, $z_3 = \dot{x}_1$, and $z_4 = \dot{x}_2$. Then to

$$[B] = \begin{bmatrix} -1000 & 400 & 0 & 0 \\ 400 & -2000 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[A] = \begin{bmatrix} 0 & 0 & -1000 & 400 \\ 0 & 0 & 400 & -2000 \\ -1000 & 400 & 0 & 0 \\ 400 & -2000 & 0 & 0 \end{bmatrix}$$

we apply [w, e] = `eig`(A, B), which yields

$$v = \begin{bmatrix} .0028 & .0457i & .0457 & -.0028i \\ -.0219 & .0118i & .0118 & .0219i \\ -.1278i & .9673 & .9673i & .1278 \\ .9916i & .2494 & .2494i & -.9916 \end{bmatrix}$$

$$d = \begin{bmatrix} -45.2942i & 0 & 0 & 0 \\ 00 & -21.1764i & 0 & 0 \\ 0 & 0 & 21.1764i & 0 \\ 0 & 0 & 0 & 45.2942i \end{bmatrix}$$

You can see that the natural frequencies are the same (21.1764 and 45.2942 rad/s) and that the eigenvectors have the same entries (but rearranged owing to the difference in the order of the x_1, x_2, $\dot{x}_1$, $\dot{x}_2$ within them). We'll discuss the details of how the eigenvectors and natural frequencies are displayed in MATLAB in the following pages.